

Mehdi Rahmani-Andebili

Precalculus

Practice Problems, Methods, and Solutions

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Preface

Calculus is one of the most important courses of many majors, including engineering and science, and even some of the non-engineering majors like economics and business, which is taught in the first semester at universities and colleges all over the world. Moreover, in many universities and colleges, precalculus is a mandatory course for the under-prepared students as the prerequisite course of calculus.

Unfortunately, some students do not have a solid background and knowledge in math and calculus when they begin their education in universities or colleges. This issue prevents them from learning other important courses like physics and calculus-based courses. Sometimes, the problem is so escalated that they give up and leave university or college. Based on my real professorship experience, students do not have a serious issue to comprehend physics and engineering courses. In fact, it is the lack of enough knowledge of calculus that hinders their understanding of calculus-based courses. Therefore, this textbook, along with the other one (*Calculus: Practice Problems, Methods, and Solutions*, Springer, 2020), has been prepared to help students succeed in their major.

Both textbooks include basic and advanced problems of calculus with very detailed problem solutions. They can be used as practicing textbooks by students and as a supplementary teaching source by instructors. Since the problems have very detailed solutions, the textbooks are useful for under-prepared students. In addition, the textbooks are beneficial for knowledgeable students because they include advanced problems. In the preparation of problem solutions, effort has been made to use typical methods to present the textbooks as an instructor-recommended one. By considering this key point, both textbooks are in the direction of instructors' lectures, and the instructors will not see any untaught and unusual problem solutions in their students' answer sheets.

To help students study the textbooks in the most efficient way, the problems have been categorized in nine different levels. In this regard, for each problem, a difficulty level (easy, normal, or hard) and a calculation amount (small, normal, or large) have been assigned. Moreover, in each chapter, problems have been ordered from the easiest one with the smallest amount of calculations to the most difficult one with the largest amount of calculations. Therefore, students are advised to study the textbooks from the easiest problem and continue practicing until they reach the normal and then the hardest one. On the other hand, this classification can help instructors choose their desirable problems to conduct a quiz or a test. Moreover, the classification of computation amount can help students manage their time during future exams and instructors assign appropriate problems based on the exam duration.

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Problems: Real Number Systems, Exponents and Radicals, and Absolute Values and Inequalities

1

Abstract

In this chapter, the basic and advanced problems of real number systems, exponents, radicals, absolute values, and inequalities are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

1.1 Real Number Systems

1.1. Which one of the numbers below exists?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) The minimum integer number smaller than -1
- 2) The minimum irrational number larger than -1
- 3) The maximum integer number smaller than -1
- 4) The maximum rational number smaller than -1

1.2. As we know, $\mathbb{R}$ is set of real numbers, $\mathbb{Z}$ is set of integer numbers, and $\mathbb{N}$ is set of natural numbers. Which one of the choices is correct?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R}$
- 2) $\mathbb{R} \subset \mathbb{Z} \subset \mathbb{N}$
- 3) $\mathbb{R} \subset \mathbb{N} \subset \mathbb{Z}$
- 4) $\mathbb{Z} \subset \mathbb{R} \subset \mathbb{N}$

1.3. If $\frac{a}{b} = \frac{c}{d}$, then which one of the choices is correct?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{b}{c} = \frac{a}{d}$
- 2) $\frac{b}{a} \neq \frac{d}{c}$
- 3) $\frac{a}{c} \neq \frac{b}{d}$
- 4) $\frac{b}{a} = \frac{d}{c}$

1.4. Which one of the choices has the greatest absolute value?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $-1 + \sqrt{2} - \sqrt{3}$

2) $-1 - \sqrt{2} + \sqrt{3}$

3) $-1 - \sqrt{2} - \sqrt{3}$

4) $1 + \sqrt{2} - \sqrt{3}$

1.5. Which one of the choices below might be a rational number if $\sqrt{\alpha}$ and β^2 are rational and irrational numbers, respectively?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $\alpha + \beta^4$

2) $\alpha + \beta$

3) $\frac{\beta}{1+\sqrt{\alpha}}$

4) $\alpha^2 + \beta$

1.2 Exponents and Radicals

1.6. Calculate the value of x^2 if $x = \sqrt[3]{2\sqrt{2}}$.

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $\sqrt{2}$

2) $\sqrt[3]{2}$

3) $\sqrt[3]{4}$

4) 2

1.7. Calculate the value of $\left(\frac{1}{\sqrt{2}} - 1\right)^{-1}$.

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $-1 + \sqrt{2}$

2) $-2 + \sqrt{2}$

3) $-(1 + \sqrt{2})$

4) $-(2 + \sqrt{2})$

1.8. Simplify and calculate the final answer of $\sqrt[3]{(-x)^3} + \sqrt{x^2} + \sqrt{(-2)^2}$ if $x > 0$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $-2x - 2$

2) -2

3) $2x + 2$

4) 2

1.9. Calculate the final answer of $\left(-\sqrt[10]{3^6}\right)^{\frac{5}{3}}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 (continued)

Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -9
- 2) 9
- 3) -3
- 4) 3

1.10. Calculate the value of $2\sqrt[3]{x^3} + \sqrt[4]{x^4}$ if $x < 0$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $3x$
- 2) x
- 3) $-x$
- 4) $-3x$

1.11. In the equation below, determine the value of x :

$$3^{x-1} = \left(\frac{1}{81}\right)^{-8}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 24
- 2) 27
- 3) 31
- 4) 33

1.12. Calculate the value of $2^{2^k} + 1$ for $k = 3$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 33
- 2) 65
- 3) 129
- 4) 257

1.13. Solve the equation below:

$$\left(\frac{3}{7}\right)^{3x-7} = \left(\frac{7}{3}\right)^{7x-3}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -1
- 2) 1
- 3) $\frac{3}{7}$
- 4) $\frac{7}{3}$

1.14. Calculate the value of the following term:

$$\left(\frac{8}{25}\right)^{-3} \times (0.8)^4 \times 0.2$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{2}{5}$
- 2) 2
- 3) $\frac{5}{2}$

4) 5

1.15. Calculate the value of the term below:

$$\frac{25}{90} \times \left(\frac{3}{2}\right)^5 \times (0.75)^{-3}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{5}{2}$
 2) $\frac{10}{3}$
 3) 5
 4) $\frac{15}{2}$

1.16. Calculate the value of the term below:

$$16 \times \left(\frac{\sqrt{2}}{2}\right)^6 \times (0.5)^{-6} \times 8^{-\frac{4}{3}}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{1}{8}$
 2) 2
 3) 8
 4) 4

1.17. Calculate the value of the term below:

$$\frac{3 \times (45)^6}{(15)^6 \times 3^7}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 1
 2) 3
 3) 5
 4) 15

1.18. Calculate the value of x if we have the following term:

$$\sqrt[5]{2} = \left(\left(\left((16)^{\frac{1}{3}} \right)^{\frac{1}{2}} \right)^{\frac{1}{3}} \right)^{\frac{1}{2}}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 5
 2) 6
 3) 9
 4) 12

1.19. Calculate the value of $(x + x^{-1})^{\frac{1}{3}}$ if $x = 1 - \sqrt{2}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $-\sqrt{2}$
 2) -1

- 3) $\sqrt[3]{2}$
 4) 1

1.20. Simplify and calculate the final answer of the term below:

$$\frac{1}{\sqrt{4} + \sqrt{11}} + \frac{1}{\sqrt{11} + \sqrt{18}} + \frac{1}{\sqrt{18} + \sqrt{25}}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{2}{7}$
 2) $\frac{3}{7}$
 3) $\frac{1}{3}$
 4) $\frac{2}{3}$

1.21. Determine the relation between A and B if:

$$A = x^{\frac{t+1}{t}}, B = x^{\frac{1}{t+1}}, t \neq 0, -1$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $A^{\frac{t}{t+1}} = B^{\frac{t+1}{t}}$
 2) $A^{\frac{t}{t+1}} = B^{t+1}$
 3) $A^{\frac{1}{t+1}} = B^{t+1}$
 4) $A^{t+1} = B^{\frac{1}{t+1}}$

1.22. Assume that a and b have different signs and $a < b$. Then, which one of the following statements is true?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $a^2 < b^2$
 2) $a^3 < b^3$
 3) $b^2 < a^3$
 4) $b^3 < a^3$

1.23. Determine the final answer of $\sqrt{4 - 2\sqrt{2}} \times \sqrt[4]{6 + 4\sqrt{2}}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\sqrt{2}$
 2) 2
 3) $2\sqrt{2}$
 4) 4

1.24. Calculate the value of $\sqrt[4]{7 - 4\sqrt{3}}\sqrt{2 + \sqrt{3}}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{1}{2}$
 2) 1
 3) $\frac{3}{2}$
 4) 2

1.25. Which one of the numbers is greater?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $1 + \sqrt[3]{2}$
- 2) $1 + \sqrt[6]{6}$
- 3) $1 + \sqrt[4]{4}$
- 4) $1 + \sqrt[3]{3}$

1.26. Determine the final answer of the term below:

$$\frac{4^{0.75}}{1 + \sqrt{2} + \sqrt{3}} + 9^{0.25}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $\sqrt{2} - 1$
- 2) 1
- 3) $\sqrt{2}$
- 4) $1 + \sqrt{2}$

1.3 Absolute Values and Inequalities

1.27. If $|-x + 1| < 2$, determine the range of x .

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $-3 < x < -1$
- 2) $-3 < x < 1$
- 3) $-1 < x < 3$
- 4) $1 < x < 3$

1.28. What is the solution of $-1 \leq 3x - 2 \leq 1$?

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\frac{1}{3} \leq x \leq 1$
- 2) $-1 \leq x \leq 1$
- 3) $-1 \leq x \leq \frac{1}{3}$
- 4) $-2 \leq x \leq 1$

1.29. Which one of the choices shows the range of the solution of $|2x - 3| \leq 5$?

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $[2, 6]$
- 2) $[-1, 4]$
- 3) $[-2, 2]$
- 4) $[-3, 1]$

- 1.30. If the inequalities of $|x - 1| < 0.1$ and $A < 2x - 3 < B$ are equivalent, calculate the value of $A + B$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -2.1
 2) -2
 3) -1.1
 4) -1

- 1.31. Which one of the choices is equivalent to $|2x - 3| < x$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $|x - 2| < 1$
 2) $|x - 1| < 2$
 3) $|x - 3| < 1$
 4) $|x - 2| < 2$

- 1.32. Solve the inequality below:

$$\frac{1}{x-1} > \frac{1}{x-3}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x < 3$
 2) $1 < x < 3$
 3) $2 < x < 3$
 4) $-2 < x < 3$

- 1.33. Solve the inequality of $|x - 1| \geq |x - 3|$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $|x| \leq 2$
 2) $|x - 2| \leq 1$
 3) $x \leq 2$
 4) $x \geq 2$

- 1.34. Solve the inequality of $|x + 1| \geq |x - 1|$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x \leq 0$
 2) $x \geq 0$
 3) $x \geq 1$
 4) $x \leq -1$

- 1.35. Which one of the choices is equivalent to $-5 < x - 11 < 3$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $|x + 5| < 2$
 2) $|x| < 7$
 3) $|x - 9| < 5$

4) $|x - 10| < 4$

1.36. Calculate the value of $|2x - 1| + |2 - x|$ if $-1 < x < 0$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $-3 - 3x$
- 2) $3 - 3x$
- 3) $-3 + 3x$
- 4) $1 + x$

1.37. Determine the neighborhood radius for the deleted symmetric neighborhood of $(3a - 7, a + 5) - \{3\}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

1.38. If $b < 0 < a$ and $|a| > |b|$, calculate the final answer of $|a + b| + |a| + |b|$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $-2b$
- 2) $-2a$
- 3) $2a$
- 4) $2b$

1.39. Determine the number of answers of x in the equation of $|x + 1| + |x - 3| = 2$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 0
- 2) 1
- 3) 2
- 4) 3

Solutions of Problems: Real Number Systems, Exponents and Radicals, and Absolute Values and Inequalities

2

Abstract

In this chapter, the problems of the first chapter are fully solved, in detail, step-by-step, and with different methods.

2.1 Real Number Systems

- 2.1. Choice (1) is not correct because set of integer numbers is not left-bounded.
 Choice (2) is not correct because between any pair of real numbers, infinite irrational numbers exist.
 Choice (3) is correct because -2 is the maximum integer number smaller than -1 .
 Choice (4) is not correct because between any pair of real numbers, infinite rational numbers exist.
 Choice (3) is the answer.
- 2.2. Figure 2.1 shows the diagram of set of real numbers, irrational numbers, rational numbers, integer numbers, whole numbers, and natural numbers. Choice (1) is the answer.

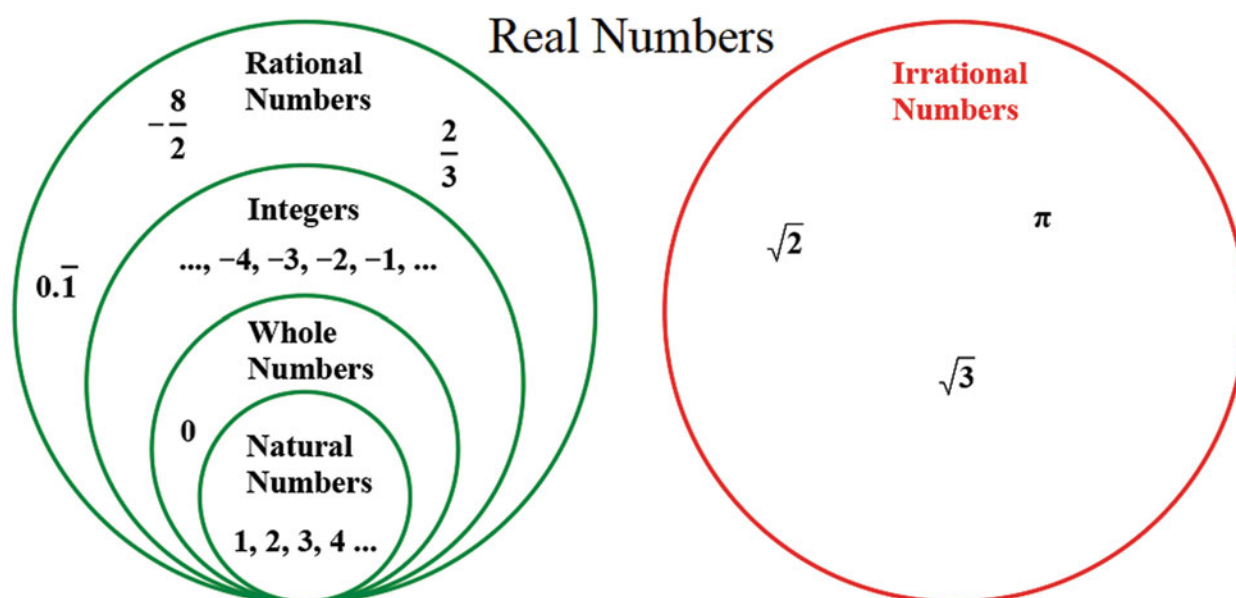


Figure 2.1 The diagram of problem 2.2

2.3. We know that:

$$\alpha = \beta \Rightarrow \frac{1}{\alpha} = \frac{1}{\beta}$$

Therefore:

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$$

Choice (4) is the answer.

2.4. As we know, $\sqrt{3} = 1.73$ and $\sqrt{2} = 1.41$, therefore:

Choice (1):

$$-1 + \sqrt{2} - \sqrt{3} < 0 \Rightarrow |-1 + \sqrt{2} - \sqrt{3}| = -(-1 + \sqrt{2} - \sqrt{3}) \approx 1.31$$

Choice (2):

$$-1 - \sqrt{2} + \sqrt{3} < 0 \Rightarrow |-1 - \sqrt{2} + \sqrt{3}| = -(-1 - \sqrt{2} + \sqrt{3}) \approx 0.68$$

Choice (3):

$$-1 - \sqrt{2} - \sqrt{3} < 0 \Rightarrow |-1 - \sqrt{2} - \sqrt{3}| = -(-1 - \sqrt{2} - \sqrt{3}) \approx 4.14$$

Choice (4):

$$1 + \sqrt{2} - \sqrt{3} > 0 \Rightarrow |1 + \sqrt{2} - \sqrt{3}| = 1 + \sqrt{2} - \sqrt{3} \approx 0.68$$

Choice (3) is the answer.

2.5. To solve the problem, we need to recall the following points:

Point 1: A rational number will remain a rational number even if it is squared.

Point 2: An irrational number might be changed to a rational number if it is squared (e.g., $\beta = \sqrt[4]{2} \Rightarrow \beta^4 = 2$).

Point 3: An irrational number will remain an irrational number if its square root is taken.

Point 4: The sum of two rational numbers is always a rational number.

Point 5: The sum of a rational number and an irrational number is always an irrational number.

Point 6: The product of a nonzero rational number and an irrational number is always an irrational number.

Point 7: A rational number will remain a rational number if it is inversed.

Choice (1) might be a rational number based on points 1, 2, and 4.

Choice (2) is an irrational number based on points 1, 3, and 5.

Choice (3) is an irrational number based on points 3, 4, 7, and 6.

Choice (4) is an irrational number based on points 1, 3, and 5.

Choice (1) is the answer.

2.2 Exponents and Radicals

2.6. We know that:

$$\sqrt[m]{a^n} = a^{\frac{n}{m}}$$

$$(a^m)^n = a^{mn}$$

Hence:

$$x = \sqrt[3]{2\sqrt{2}} = \left(2^{\frac{3}{2}}\right)^{\frac{1}{3}} = 2^{\frac{1}{2}} \Rightarrow x^2 = \left(2^{\frac{1}{2}}\right)^2 = 2$$

Choice (4) is the answer.

2.7. We know that:

$$(a + b)(a - b) = a^2 - b^2$$

Thus:

$$\left(\frac{1}{\sqrt{2}} - 1\right)^{-1} = \frac{1}{\frac{1}{\sqrt{2}} - 1} = \frac{\sqrt{2}}{1 - \sqrt{2}} = \frac{\sqrt{2}}{1 - \sqrt{2}} \times \frac{1 + \sqrt{2}}{1 + \sqrt{2}} = \frac{\sqrt{2}(1 + \sqrt{2})}{1 - 2} = -(\sqrt{2} + 2)$$

Choice (4) is the answer.

2.8. From exponent and radical rules, we know that:

$$\sqrt[n]{a^n} = a, \quad \text{if } n \text{ is an odd number}$$

$$\sqrt[n]{a^n} = |a|, \quad \text{if } n \text{ is an even number}$$

Therefore:

$$\sqrt[3]{(-x)^3} + \sqrt{x^2} + \sqrt{(-2)^2} = (-x) + |x| + |-2|$$

$$\xrightarrow{x > 0} -x + x + 2 = 2$$

Choice (4) is the answer.

2.9. From exponent and radical rules, we know that:

$$(ab)^n = a^n b^n$$

$$(a^m)^n = a^{mn}$$

Hence:

$$\left(-\sqrt[10]{3^6}\right)^{\frac{5}{3}} = \left(-1 \times 3^{\frac{6}{10}}\right)^{\frac{5}{3}} = (-1)^{\frac{5}{3}} \left(3^{\frac{6}{10}}\right)^{\frac{5}{3}} = \left((-1)^5\right)^{\frac{1}{3}} \left(3^{\frac{6}{10} \times \frac{5}{3}}\right) = (-1)^{\frac{1}{3}} (3^1) = -1 \times 3 = -3$$

Choice (3) is the answer.

2.10. From exponent and radical rules, we know that:

$$\sqrt[n]{a^n} = a, \quad \text{if } n \text{ is an odd number}$$

$$\sqrt[n]{a^n} = |a|, \quad \text{if } n \text{ is an even number}$$

Therefore:

$$2\sqrt[3]{x^3} + \sqrt[4]{x^4} = 2x + |x|$$

$$\xrightarrow{x > 0} 2x - x = x$$

Choice (2) is the answer.

2.11. We know that:

$$\left(\frac{1}{a}\right)^{-n} = a^n$$

Therefore:

$$\left(\frac{1}{81}\right)^{-8} = 81^8 = (3^4)^8 = 3^{32}$$

$$3^{x-1} = 3^{32} \Rightarrow x - 1 = 32 \Rightarrow x = 33$$

Choice (4) is the answer.

2.12. From exponent rules, we know that:

$$a^{n^m} = a^{\overbrace{n \times \dots \times n}^{m \text{ times}}}$$

Therefore for $k = 3$, we can write:

$$2^{2^k} + 1 = 2^{2^3} + 1 = 2^8 + 1 = 256 + 1 = 257$$

Choice (4) is the answer.

2.13. From exponent rules, we know that:

$$a^n = \left(\frac{1}{a}\right)^{-n}$$

$$a^n = a^m \Rightarrow n = m$$

Therefore:

$$\left(\frac{3}{7}\right)^{3x-7} = \left(\frac{7}{3}\right)^{7x-3} \Rightarrow \left(\frac{3}{7}\right)^{3x-7} = \left(\frac{3}{7}\right)^{-7x+3}$$

$$\Rightarrow 3x - 7 = -7x + 3 \Rightarrow 10x = 10 \Rightarrow x = 1$$

Choice (2) is the answer.

2.14. From exponent rules, we know that:

$$a^n = \left(\frac{1}{a}\right)^{-n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^m)^n = a^{mn}$$

$$\frac{a^n}{a^m} = a^{n-m}$$

The problem can be solved as follows:

$$\left(\frac{8}{25}\right)^{-3} = \left(\frac{25}{8}\right)^3 = \frac{(5^2)^3}{(2^3)^3} = \frac{5^6}{2^9}$$

$$(0.8)^4 = \left(\frac{8}{10}\right)^4 = \frac{(2^3)^4}{(2 \times 5)^4} = \frac{2^{12}}{2^4 \times 5^4} = \frac{2^8}{5^4}$$

$$0.2 = \frac{2}{10} = \frac{1}{5}$$

Therefore:

$$\left(\frac{8}{25}\right)^{-3} \times (0.8)^4 \times 0.2 = \frac{5^6}{2^9} \times \frac{2^8}{5^4} \times \frac{1}{5} = \frac{5^6}{5^5} \times \frac{2^8}{2^9} = 5 \times \frac{1}{2} = \frac{5}{2}$$

Choice (3) is the answer.

2.15. From exponent rules, we know that:

$$a^n = \left(\frac{1}{a}\right)^{-n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^m)^n = a^{mn}$$

$$\frac{a^n}{a^m} = a^{n-m}$$

Therefore:

$$\begin{aligned} \frac{25}{90} \times \left(\frac{3}{2}\right)^5 \times (0.75)^{-3} &= \frac{5}{18} \times \left(\frac{3}{2}\right)^5 \times \left(\frac{3}{4}\right)^{-3} = \frac{5}{2 \times 3^2} \times \frac{3^5}{2^5} \times \frac{4^3}{3^3} = \frac{5}{2 \times 3^2} \times \frac{3^5}{2^5} \times \frac{(2^2)^3}{3^3} \\ &= 5 \times \frac{3^5}{3^2 \times 3^3} \times \frac{2^6}{2 \times 2^5} = 5 \times \frac{3^5}{3^5} \times \frac{2^6}{2^6} = 5 \times 1 \times 1 = 5 \end{aligned}$$

Choice (3) is the answer.

2.16. From exponent rules, we know that:

$$a^n = \left(\frac{1}{a}\right)^{-n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^m)^n = a^{mn}$$

$$\frac{a^n}{a^m} = a^{n-m}$$

Hence:

$$\begin{aligned} 16 \times \left(\frac{\sqrt{2}}{2}\right)^6 \times (0.5)^{-6} \times 8^{-\frac{4}{3}} &= 2^4 \times \left(\frac{1}{2^{\frac{1}{2}}}\right)^6 \times \left(\frac{1}{2}\right)^{-6} \times (2^3)^{-\frac{4}{3}} \\ &= 2^4 \times \frac{1}{2^3} \times 2^6 \times \frac{1}{2^4} = \frac{2^4 \times 2^6}{2^4 \times 2^3} = 2^{4+6-4-3} = 2^3 = 8 \end{aligned}$$

Choice (3) is the answer.

2.17. From exponent rules, we know that:

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

Thus:

$$\frac{3 \times (45)^6}{(15)^6 \times 3^7} = \frac{3 \times (5 \times 3^2)^6}{(5 \times 3)^6 \times 3^7} = \frac{3 \times 5^6 \times 3^{12}}{5^6 \times 3^6 \times 3^7} = \frac{3^{1+12}}{3^{6+7}} = 1$$

Choice (1) is the answer.

2.18. From exponent rules, we know that:

$$\sqrt[m]{2} = 2^{\frac{1}{m}}$$

$$(a^m)^n = a^{mn}$$

$$a^n = a^m \Rightarrow n = m$$

Hence:

$$\sqrt[x]{2} = 2^{\frac{1}{x}}$$

$$\left(\left(\left((16)^{\frac{1}{3}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} = \left(\left(\left((2^4)^{\frac{1}{3}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} = 2^{4 \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}} = 2^{\frac{1}{6}}$$

Therefore:

$$\sqrt[x]{2} = \left(\left(\left((16)^{\frac{1}{3}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} \right)^{\frac{1}{2}} \Rightarrow 2^{\frac{1}{x}} = 2^{\frac{1}{6}} \Rightarrow x = 6$$

Choice (2) is the answer.

2.19. We know that:

$$(a+b)(a-b) = a^2 - b^2$$

$$(a^m)^n = a^{mn}$$

Based on the information given in the problem, we have:

$$x = 1 - \sqrt{2}$$

Therefore:

$$\begin{aligned} (x + x^{-1})^{\frac{1}{3}} &= \left(1 - \sqrt{2} + \frac{1}{1 - \sqrt{2}} \right)^{\frac{1}{3}} = \left(1 - \sqrt{2} + \frac{1}{1 - \sqrt{2}} \times \frac{1 + \sqrt{2}}{1 + \sqrt{2}} \right)^{\frac{1}{3}} = \left(1 - \sqrt{2} + \frac{1 + \sqrt{2}}{1 - 2} \right)^{\frac{1}{3}} \\ &= (1 - \sqrt{2} - 1 - \sqrt{2})^{\frac{1}{3}} = (-2\sqrt{2})^{\frac{1}{3}} = \left((-\sqrt{2})^3 \right)^{\frac{1}{3}} = -\sqrt{2} \end{aligned}$$

Choice (1) is the answer.

2.20. Based on the rule of difference of two squares, we know that:

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

To solve this problem, we need to change the denominator of each fraction to a rational number as follows:

$$\begin{aligned} & \frac{1}{\sqrt{4} + \sqrt{11}} + \frac{1}{\sqrt{11} + \sqrt{18}} + \frac{1}{\sqrt{18} + \sqrt{25}} \\ &= \frac{1}{\sqrt{4} + \sqrt{11}} \times \frac{\sqrt{4} - \sqrt{11}}{\sqrt{4} - \sqrt{11}} + \frac{1}{\sqrt{11} + \sqrt{18}} \times \frac{\sqrt{11} - \sqrt{18}}{\sqrt{11} - \sqrt{18}} + \frac{1}{\sqrt{18} + \sqrt{25}} \times \frac{\sqrt{18} - \sqrt{25}}{\sqrt{18} - \sqrt{25}} \\ &= \frac{\sqrt{4} - \sqrt{11}}{4 - 11} + \frac{\sqrt{11} - \sqrt{18}}{11 - 18} + \frac{\sqrt{18} - \sqrt{25}}{18 - 25} = \frac{\sqrt{4} - \sqrt{11} + \sqrt{11} - \sqrt{18} + \sqrt{18} - \sqrt{25}}{-7} \\ &= \frac{\sqrt{4} - \sqrt{25}}{-7} = \frac{2 - 5}{-7} = \frac{3}{7} \end{aligned}$$

Choice (2) is the answer.

2.21. From the problem, we have:

$$A = x^{\frac{t+1}{t}}, B = x^{\frac{1}{t+1}}, t \neq 0, -1$$

From exponent rules, we know that:

$$(a^m)^n = a^{mn}$$

In choice (1):

$$\begin{cases} A^{\frac{1}{t+1}} = \left(x^{\frac{t+1}{t}}\right)^{\frac{1}{t+1}} = x \\ B^{\frac{t+1}{t}} = \left(x^{\frac{1}{t+1}}\right)^{\frac{t+1}{t}} = x^{\frac{1}{t}} \end{cases} \Rightarrow A^{\frac{1}{t+1}} \neq B^{\frac{t+1}{t}}$$

In choice (2):

$$\begin{cases} A^{\frac{1}{t+1}} = \left(x^{\frac{t+1}{t}}\right)^{\frac{1}{t+1}} = x \\ B^{t+1} = \left(x^{\frac{1}{t+1}}\right)^{t+1} = x \end{cases} \Rightarrow A^{\frac{1}{t+1}} = B^{t+1}$$

In choice (3):

$$\begin{cases} A^{\frac{1}{t+1}} = \left(x^{\frac{t+1}{t}}\right)^{\frac{1}{t+1}} = x^{\frac{1}{t}} \\ B^{t+1} = \left(x^{\frac{1}{t+1}}\right)^{t+1} = x \end{cases} \Rightarrow A^{\frac{1}{t+1}} \neq B^{t+1}$$

In choice (4):

$$\begin{cases} A^{t+1} = \left(x^{\frac{t+1}{t}}\right)^{t+1} = x^{\frac{(t+1)^2}{t}} \\ B^{\frac{1}{t+1}} = \left(x^{\frac{1}{t+1}}\right)^{\frac{1}{t+1}} = x^{\frac{1}{(t+1)^2}} \end{cases} \Rightarrow A^{t+1} \neq B^{\frac{1}{t+1}}$$

Choice (2) is the answer.

2.22. Based on the information given in the problem, we know that a and b have different signs and $a < b$. Therefore:

$$a < 0, b > 0$$

$$\stackrel{(\cdot)^3}{\Rightarrow} a^3 < 0, b^3 > 0 \Rightarrow a^3 < b^3$$

Choice (2) is the answer.

2.23. We know that:

$$(a + b)(a - b) = a^2 - b^2$$

$$\sqrt[m]{a^n} = a^{\frac{n}{m}}$$

The problem can be solved as follows:

$$\begin{aligned} \sqrt{4 - 2\sqrt{2}} \times \sqrt[4]{6 + 4\sqrt{2}} &= \sqrt{4 - 2\sqrt{2}} \times \sqrt[4]{2^2 + (\sqrt{2})^2 + 4\sqrt{2}} = \sqrt{2(2 - \sqrt{2})} \times \sqrt[4]{(2 + \sqrt{2})^2} \\ &= \sqrt{2(2 - \sqrt{2})} \times \sqrt{2 + \sqrt{2}} = \sqrt{2(2 - \sqrt{2})(2 + \sqrt{2})} = \sqrt{2(4 - 2)} = \sqrt{4} = 2 \end{aligned}$$

Choice (2) is the answer.

2.24. We know that:

$$(a + b)(a - b) = a^2 - b^2$$

$$\sqrt[m]{a^n} = a^{\frac{n}{m}}$$

Therefore:

$$\begin{aligned} \sqrt[4]{7 - 4\sqrt{3}} \sqrt{2 + \sqrt{3}} &= \sqrt[4]{7 - 4\sqrt{3}} \sqrt[4]{(2 + \sqrt{3})^2} = \sqrt[4]{(7 - 4\sqrt{3})(2 + \sqrt{3})^2} = \sqrt[4]{(7 - 4\sqrt{3})(4 + 4\sqrt{3} + 3)} \\ &= \sqrt[4]{(7 - 4\sqrt{3})(7 + 4\sqrt{3})} = \sqrt[4]{(7)^2 - (4\sqrt{3})^2} = \sqrt[4]{49 - 48} = \sqrt[4]{1} = 1 \end{aligned}$$

Choice (2) is the answer.

2.25. From radical rules, we know that:

$$\sqrt[m]{a} = \sqrt[m \times n]{a^n}$$

Now, we should define the radicals based on a common root (greatest common divisor (GCL)), as follows:

$$GCL\{2, 6, 4, 3\} = 12$$

Therefore:

$$1 + \sqrt[2]{2} = 1 + \sqrt[2 \times 6]{2^6} = 1 + \sqrt[12]{64}$$

$$1 + \sqrt[6]{6} = 1 + \sqrt[6 \times 2]{6^2} = 1 + \sqrt[12]{36}$$

$$1 + \sqrt[4]{4} = 1 + \sqrt[4 \times 3]{4^3} = 1 + \sqrt[12]{64}$$

$$1 + \sqrt[3]{3} = 1 + \sqrt[3 \times 4]{3^4} = 1 + \sqrt[12]{81}$$

As can be seen, $1 + \sqrt[3]{3}$ is the greatest number. **Choice (4) is the answer.**

2.26. We know that:

$$(a + b)(a - b) = a^2 - b^2$$

$$(a^m)^n = a^{mn}$$

The problem can be solved as follows:

$$\begin{aligned} \frac{4^{0.75}}{1 + \sqrt{2} + \sqrt{3}} + 9^{0.25} &= \frac{(2^2)^{\frac{3}{4}}}{1 + \sqrt{2} + \sqrt{3}} + (3^2)^{\frac{1}{4}} = \frac{2^{\frac{3}{2}}}{1 + \sqrt{2} + \sqrt{3}} + 3^{\frac{1}{2}} = \frac{2\sqrt{2}}{1 + \sqrt{2} + \sqrt{3}} + \sqrt{3} \\ &= \frac{2\sqrt{2}}{1 + \sqrt{2} + \sqrt{3}} \times \frac{1 + \sqrt{2} - \sqrt{3}}{1 + \sqrt{2} - \sqrt{3}} + \sqrt{3} = \frac{2\sqrt{2}(1 + \sqrt{2} - \sqrt{3})}{(1 + \sqrt{2})^2 - (\sqrt{3})^2} + \sqrt{3} = \frac{2\sqrt{2}(1 + \sqrt{2} - \sqrt{3})}{1 + 2\sqrt{2} + 2 - 3} + \sqrt{3} \\ &= \frac{2\sqrt{2}(1 + \sqrt{2} - \sqrt{3})}{2\sqrt{2}} + \sqrt{3} = 1 + \sqrt{2} - \sqrt{3} + \sqrt{3} = 1 + \sqrt{2} \end{aligned}$$

Choice (4) is the answer.

2.3 Absolute Values and Inequalities

2.27. We know from the features of absolute value and inequality that:

$$|x - a| < b \Leftrightarrow -b < x - a < b$$

The problem can be solved as follows:

$$|-x + 1| < 2 \Rightarrow -2 < -x + 1 < 2 \xrightarrow{-1} -3 < -x < 1 \xrightarrow{\times(-1)} -1 < x < 3$$

Choice (3) is the answer.

2.28. The problem can be solved as follows:

$$-1 \leq 3x - 2 \leq 1 \xrightarrow{+2} 1 \leq 3x \leq 3 \xrightarrow{\times \frac{1}{3}} \frac{1}{3} \leq x \leq 1$$

Choice (1) is the answer.

2.29. We know from the features of absolute value and inequality that:

$$|x - a| < b \Leftrightarrow -b < x - a < b$$

The problem can be solved as follows:

$$|2x - 3| \leq 5 \Rightarrow -5 \leq 2x - 3 \leq 5 \xrightarrow{+3} -2 \leq 2x \leq 8 \xrightarrow{\times \frac{1}{2}} -1 \leq x \leq 4$$

Choice (2) is the answer.

2.30. We know from the features of absolute value and inequality that:

$$|x - a| < b \Leftrightarrow -b < x - a < b$$

Based on the information given in the problem, we have:

$$|x - 1| < 0.1 \Leftrightarrow A < 2x - 3 < B \quad (1)$$

The problem can be solved as follows:

$$|x - 1| < 0.1 \Rightarrow -0.1 < x - 1 < 0.1 \xrightarrow{\times 2} -0.2 < 2x - 2 < 0.2 \xrightarrow{+1} -1.2 < 2x - 3 < -0.8 \quad (2)$$

By comparing (1) and (2), we can conclude that:

$$A = -1.2, B = -0.8 \Rightarrow A + B = -2$$

Choice (2) is the answer.

2.31. We know from the features of absolute value and inequality that:

$$|x - a| < b \Leftrightarrow -b < x - a < b$$

The problem can be solved as follows:

$$|2x - 3| < x \Rightarrow -x < 2x - 3 < x \Rightarrow \begin{cases} 2x - 3 < x \Rightarrow x < 3 \\ -x < 2x - 3 \Rightarrow 1 < x \end{cases} \Rightarrow 1 < x < 3 \xrightarrow{-2} -1 < x - 2 < 1 \Rightarrow |x - 2| < 1$$

Choice (1) is the answer.

2.32. The problem can be solved as follows:

$$\frac{1}{x-1} > \frac{1}{x-3} \Rightarrow \frac{1}{x-1} - \frac{1}{x-3} > 0 \Rightarrow \frac{(x-3) - (x-1)}{(x-1)(x-3)} > 0 \Rightarrow \frac{-2}{(x-1)(x-3)} > 0$$

$$\Rightarrow (x-1)(x-3) < 0 \Rightarrow 1 < x < 3$$

Choice (2) is the answer.

2.33. The problem can be solved as follows:

$$|x - 1| \geq |x - 3| \xrightarrow{(\cdot)^2} (x - 1)^2 \geq (x - 3)^2 \Rightarrow x^2 - 2x + 1 \geq x^2 - 6x + 9 \Rightarrow 4x \geq 8 \Rightarrow x \geq 2$$

Choice (4) is the answer.

2.34. The problem can be solved as follows:

$$|x + 1| \geq |x - 1| \xrightarrow{(\cdot)^2} (x + 1)^2 \geq (x - 1)^2 \Rightarrow x^2 + 2x + 1 \geq x^2 - 2x + 1 \Rightarrow 4x \geq 0 \Rightarrow x \geq 0$$

Choice (2) is the answer.

2.35. We know from the features of absolute value and inequality that:

$$|x - a| < b \Leftrightarrow -b < x - a < b$$

The problem can be solved as follows:

$$-5 < x - 11 < 3 \xrightarrow{+11} 6 < x < 14 \xrightarrow{-10} -4 < x - 10 < 4 \Rightarrow |x - 10| < 4$$

Choice (4) is the answer.

2.36. The problem can be solved as follows:

$$-1 < x < 0 \xrightarrow{\times 2} -2 < 2x < 0 \xrightarrow{-1} -3 < 2x - 1 < -1 \Rightarrow |2x - 1| < 1 - 2x$$

$$-1 < x < 0 \xrightarrow{\times (-1)} 0 < -x < 1 \xrightarrow{+2} 2 < 2 - x < 3 \Rightarrow |2 - x| < 2 - x$$

$$\Rightarrow |2x - 1| + |2 - x| = 1 - 2x + 2 - x = 3 - 3x$$

Choice (2) is the answer.

2.37. The neighborhood center (C) and neighborhood radius (R) for the symmetric neighborhood of $(\alpha_1, \alpha_3) - \{\alpha_2\}$ can be determined as follows:

$$C = \alpha_2 = \frac{\alpha_1 + \alpha_3}{2} \quad (1)$$

$$R = \frac{\alpha_3 - \alpha_1}{2} \quad (2)$$

Therefore, for $(3a - 7, a + 5) - \{3\}$, we can write:

$$3 = \frac{(3a - 7) + (a + 5)}{2} \Rightarrow 3 = \frac{4a - 2}{2} \Rightarrow a = 2 \quad (3)$$

$$R = \frac{(a + 5) - (3a - 7)}{2} = \frac{-2a + 12}{2} \quad (4)$$

Solving (3) and (4):

$$R = \frac{-2 \times 2 + 12}{2} = 4$$

Choice (4) is the answer.

2.38. The problem can be solved as follows:

$$a > 0 \Rightarrow |a| = a \quad (1)$$

$$b < 0 \Rightarrow |b| = -b \quad (2)$$

$$|a| > |b| \xrightarrow{(1), (2)} a > -b \Rightarrow a + b > 0 \Rightarrow |a + b| = a + b \quad (3)$$

Solving (1), (2), and (3):

$$|a + b| + |a| + |b| = a + b + a + (-b) = 2a$$

Choice (3) is the answer.

2.39. From the problem, we have:

$$|x + 1| + |x - 3| = 2 \tag{1}$$

We know that:

$$|A| + |B| \geq |A - B|$$

Therefore:

$$|x + 1| + |x - 3| \geq |x + 1 - (x - 3)| \Rightarrow |x + 1| + |x - 3| \geq 4 \tag{2}$$

Solving (1) and (2):

$$2 \geq 4 \Rightarrow \text{Wrong}$$

Therefore, there is no answer for the problem. Choice (1) is the answer.

Problems: Systems of Equations

3

Abstract

In this chapter, the basic and advanced problems of systems of equations are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

3.1. For what value of m , the system of equations below has a unique solution?

$$\begin{cases} 2x - 3y = 5 \\ x + my = 2 \end{cases}$$

Difficulty level ☒ Easy ☐ Normal ☐ Hard
Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $m = \frac{3}{2}$
- 2) $m \neq -\frac{3}{2}$
- 3) $m = -\frac{3}{2}$
- 4) For any m

3.2. If the value of x from the system of equations below is 2.5, determine the value of $2a + b$:

$$\begin{cases} ax - y = 1 \\ bx + 2y = 3 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -2
- 2) -1
- 3) 1
- 4) 2

3.3. Calculate the value of $y^2 + y$ from the solution of the system of equations below:

$$\begin{cases} (x + 1)^2 + 3y = 8 \\ x(x + 2) - 5y = 31 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 12
- 2) 6
- 3) 10
- 4) 8

3.4. For what value of a , the matrix below has a unique solution?

$$\begin{bmatrix} a+1 & 2 \\ -1 & a-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ 1 \end{bmatrix}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\{-1, 1\}$
- 2) $\mathbb{R} - \{0, 1\}$
- 3) For no value of a
- 4) $\mathbb{R}$

3.5. Calculate the value of y from the solution of the system of equations below:

$$\begin{cases} \frac{x}{2} - \frac{y}{3} = \frac{4}{3} \\ 2x + 3y = 14 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -4
- 2) 2
- 3) 3
- 4) 4

3.6. Solve the system of equations below:

$$\begin{cases} \frac{x-1}{2} = \frac{y-1}{3} \\ x + 2y = 11 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x = -3, y = -4$
- 2) $x = 3, y = 4$
- 3) $x = 3, y = -4$
- 4) $x = -3, y = 4$

3.7. Calculate the value of $x + y$ from the solution of the system of equations below:

$$\begin{cases} x + 2y = -1 \\ 3x - y = 4 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -1
- 2) 0
- 3) 1
- 4) 2

3.8. For what value of a , the system of equations below has an infinite number of solutions?

$$\begin{cases} 2x + 3y = 4 \\ y = a(x - 2) \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $-\frac{3}{2}$
- 2) $-\frac{2}{3}$
- 3) $\frac{2}{3}$
- 4) $\frac{3}{2}$

3.9. Calculate the value of $y - x$ from the solution of the system of equations below:

$$\begin{cases} x + 3y = -2 \\ 3x + y = -14 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -4
- 2) -3
- 3) 3
- 4) 6

3.10. Calculate the value of y from the solution of the system of equations below:

$$\begin{cases} \frac{x}{y} = 2 \\ x + 2y = 4 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 4
- 2) 3
- 3) 2
- 4) 1

3.11. Calculate the value of xy from the solution of the system of equations below:

$$\begin{cases} 3x + 2y = -1 \\ 2x + 3y = -4 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -3
- 2) -2
- 3) 2
- 4) 3

3.12. Calculate the determinant of matrix of A if:

$$A^{-1} = \begin{bmatrix} 2 & 3 \\ -7 & 6 \end{bmatrix}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\frac{1}{33}$
- 2) $\frac{1}{9}$
- 3) 1
- 4) 33

3.13. Calculate the matrix X if we have:

$$AX = 2I, A = \begin{bmatrix} -2 & -1 \\ 4 & 3 \end{bmatrix}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\begin{bmatrix} -3 & -1 \\ 4 & 2 \end{bmatrix}$
- 2) $\begin{bmatrix} 3 & 1 \\ 4 & -2 \end{bmatrix}$
- 3) $2 \begin{bmatrix} -3 & -1 \\ 4 & 2 \end{bmatrix}$
- 4) $\frac{1}{2} \begin{bmatrix} 3 & 1 \\ 4 & -2 \end{bmatrix}$

3.14. Calculate the value of $x + y$:

$$\begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

3.15. Which one of the matrices is invertible?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$
- 2) $\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$
- 3) $\begin{bmatrix} 2 & 8 \\ 3 & 12 \end{bmatrix}$
- 4) $\begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix}$

3.16. Calculate the value of b if:

$$A = \begin{bmatrix} -1 & b \\ 0 & 1 \end{bmatrix}, A^{-1} = \begin{bmatrix} -1 & 3 \\ 0 & 1 \end{bmatrix}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 0
- 2) 1
- 3) 2
- 4) 3

3.17. For what value of m , the system of equations below has an infinite number of solutions?

$$\begin{cases} m(x-1) = 3(x-y) \\ 4x + (m+1)y = 2 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 2
- 2) -3
- 3) 3
- 4) 5

3.18. For what value of m , the system of equations below does not have a solution?

$$\begin{cases} my + 2x = 5 \\ y + (m-1)x = 2m-3 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -2
- 2) -1
- 3) 1
- 4) 2

3.19. Calculate the value of $x + y$ from the solution of the following system of equations:

$$\begin{cases} \frac{3}{2x-1} - \frac{1}{y-2} = \frac{4}{3} \\ \frac{1}{2x-1} + \frac{5}{2-y} = 2 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

3.20. How many solution(s) does the system of equations below have?

$$\frac{5x+y}{1} = \frac{3x-y}{3} = \frac{7x+y}{2}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

Solutions of Problems: Systems of Equations

4

Abstract

In this chapter, the problems of the third chapter are fully solved, in detail, step-by-step, and with different methods.

4.1. The system of equations with the form presented in (1) has a unique solution (the lines intersect each other) if:

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad (1)$$

Therefore:

$$\begin{cases} 2x - 3y = 5 \\ x + my = 2 \end{cases} \Rightarrow \frac{2}{1} \neq \frac{-3}{m} \Rightarrow m \neq -\frac{3}{2}$$

Choice (2) is the answer.

4.2. The problem can be solved as follows:

$$\begin{cases} ax - y = 1 \\ bx + 2y = 3 \end{cases} \xrightarrow{x=2.5} \begin{cases} 2.5a - y = 1 \\ 2.5b + 2y = 3 \end{cases} \xrightarrow{\times 2} \begin{cases} 5a - 2y = 2 \\ 2.5b + 2y = 3 \end{cases} \xrightarrow{+} \begin{cases} 5a - 2y = 2 \\ 5a + 2.5b = 5 \end{cases} \xrightarrow{\times \frac{2}{5}} \begin{cases} 5a - 2y = 2 \\ 2a + b = 2 \end{cases}$$

Choice (4) is the answer.

4.3. The problem can be solved as follows:

$$\begin{cases} (x+1)^2 + 3y = 8 \\ x(x+2) - 5y = 31 \end{cases} \Rightarrow \begin{cases} x^2 + 2x + 3y = 7 \\ x^2 + 2x - 5y = 31 \end{cases} \xrightarrow{-} \begin{cases} x^2 + 2x + 3y = 7 \\ 8y = -24 \end{cases} \Rightarrow y = -3 \Rightarrow y^2 = 9$$

$$\Rightarrow y^2 + y = 9 + (-3) = 6$$

Choice (2) is the answer.

4.4. The system of equations with the matrix form presented in (1) has a solution (the lines intersect each other) if its determinant is nonzero as follows:

$$|A| \neq 0 \Rightarrow a_1a_4 - a_2a_3 \neq 0$$

$$AX = B \Rightarrow \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad (1)$$

Therefore:

$$\begin{bmatrix} a+1 & 2 \\ -1 & a-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ 1 \end{bmatrix} \Rightarrow |A| = (a+1)(a-1) - (2)(-1) = a^2 - 1 + 2 \neq 0 \Rightarrow a^2 + 1 \neq 0 \Rightarrow a \in \mathbb{R}$$

Choice (4) is the answer.

4.5. The problem can be solved as follows:

$$\begin{cases} \frac{x}{2} - \frac{y}{3} = \frac{4}{3} \\ 2x + 3y = 14 \end{cases} \xrightarrow[\times(-3)]{\times 12} \begin{cases} 6x - 4y = 16 \\ -6x - 9y = -42 \end{cases} \xrightarrow{+} -13y = -26 \Rightarrow y = 2$$

Choice (2) is the answer.

4.6. The problem can be solved as follows:

$$\begin{cases} \frac{x-1}{2} = \frac{y-1}{3} \\ x + 2y = 11 \end{cases} \xrightarrow[\Rightarrow]{\times 6} \begin{cases} 3x - 3 = 2y - 2 \\ x + 2y = 11 \end{cases} \Rightarrow \begin{cases} 3x - 2y = 1 \\ x + 2y = 11 \end{cases} \xrightarrow{+} 4x = 12 \Rightarrow x = 3$$

$$x + 2y = 11 \xrightarrow{x=3} 3 + 2y = 11 \Rightarrow y = 4$$

Choice (2) is the answer.

4.7. The problem can be solved as follows:

$$\begin{cases} x + 2y = -1 \\ 3x - y = 4 \end{cases} \xrightarrow[\Rightarrow]{\times 2} \begin{cases} x + 2y = -1 \\ 6x - 2y = 8 \end{cases} \xrightarrow{+} 7x = 7 \Rightarrow x = 1$$

$$3x - y = 4 \xrightarrow{x=1} 3 \times 1 - y = 4 \Rightarrow y = -1$$

$$\Rightarrow x + y = 1 + (-1) = 0$$

Choice (2) is the answer.

4.8. The system of equations with the form presented in (1) has an infinite number of solutions (the lines match each other) if:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad (1)$$

Thus:

$$\begin{cases} 2x + 3y = 4 \\ y = a(x - 2) \end{cases} \Rightarrow \begin{cases} 2x + 3y = 4 \\ ax - y = 2a \end{cases} \Rightarrow \frac{2}{a} = \frac{3}{-1} = \frac{4}{2a} \Rightarrow a = -\frac{2}{3}$$

Choice (2) is the answer.

4.9. The problem can be solved as follows:

$$\begin{aligned} \begin{cases} x + 3y = -2 \\ 3x + y = -14 \end{cases} &\xrightarrow[\Rightarrow]{\times(-3)} \begin{cases} -3x - 9y = 6 \\ 3x + y = -14 \end{cases} \xrightarrow{+} \begin{cases} -8y = -8 \\ \end{cases} \Rightarrow y = 1 \\ 3x + y = -14 &\xrightarrow{y=1} 3x + 1 = -14 \Rightarrow x = -5 \\ &\Rightarrow y - x = 1 - (-5) = 6 \end{aligned}$$

Choice (4) is the answer.

4.10. The problem can be solved as follows:

$$\begin{cases} \frac{x}{y} = 2 \\ x + 2y = 4 \end{cases} \Rightarrow \begin{cases} x - 2y = 0 \\ x + 2y = 4 \end{cases} \xrightarrow[\Rightarrow]{\times(-1)} \begin{cases} -x + 2y = 0 \\ x + 2y = 4 \end{cases} \xrightarrow{+} \begin{cases} 4y = 4 \\ \end{cases} \Rightarrow y = 1$$

Choice (4) is the answer.

4.11. The problem can be solved as follows:

$$\begin{aligned} \begin{cases} 3x + 2y = -1 \\ 2x + 3y = -4 \end{cases} &\xrightarrow[\Rightarrow]{\times(-2)} \begin{cases} -6x - 4y = 2 \\ 2x + 3y = -4 \end{cases} \xrightarrow{\times 3} \begin{cases} -6x - 4y = 2 \\ 6x + 9y = -12 \end{cases} \xrightarrow{+} \begin{cases} 5y = -10 \\ \end{cases} \Rightarrow y = -2 \\ 2x + 3y = -4 &\xrightarrow{y=-2} 2x + 3 \times (-2) = -4 \Rightarrow 2x = 2 \Rightarrow x = 1 \\ &\Rightarrow xy = 1 \times (-2) = -2 \end{aligned}$$

Choice (2) is the answer.

4.12. The determinant of a matrix is determined as follows:

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix} = a_1 a_4 - a_2 a_3$$

The relation below exists between the determinants of a matrix and its inverse matrix:

$$|A^{-1}| = \frac{1}{|A|}$$

Therefore:

$$\begin{aligned} A^{-1} = \begin{bmatrix} 2 & 3 \\ -7 & 6 \end{bmatrix} &\Rightarrow |A^{-1}| = \begin{vmatrix} 2 & 3 \\ -7 & 6 \end{vmatrix} = 2 \times 6 - (-7) \times 3 = 12 + 21 = 33 \\ &\Rightarrow |A| = \frac{1}{|A^{-1}|} = \frac{1}{33} \end{aligned}$$

Choice (1) is the answer.

4.13. The system of equations in matrix form can be solved as follows:

$$AX = B \Rightarrow X = A^{-1}B$$

$$\begin{aligned} \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} &= \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}^{-1} \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} &= \frac{1}{a_1a_4 - a_2a_3} \begin{bmatrix} a_4 & -a_2 \\ -a_3 & a_1 \end{bmatrix} \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \end{aligned}$$

Therefore:

$$\begin{aligned} AX = 2I &\Rightarrow \begin{bmatrix} -2 & -1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} &= \begin{bmatrix} -2 & -1 \\ 4 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \frac{1}{(-2)(3) - (4)(-1)} \begin{bmatrix} 3 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 6 & 2 \\ -8 & -4 \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ 4 & 2 \end{bmatrix} \end{aligned}$$

Choice (1) is the answer.

4.14. The system of equations in matrix form can be solved as follows:

$$AX = B \Rightarrow X = A^{-1}B$$

$$\begin{aligned} \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \frac{1}{a_1a_4 - a_2a_3} \begin{bmatrix} a_4 & -a_2 \\ -a_3 & a_1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{aligned}$$

Therefore:

$$\begin{aligned} \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{(2)(-1) - (1)(-1)} \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ \Rightarrow x = 2, y = 1 &\Rightarrow x + y = 3 \end{aligned}$$

Choice (3) is the answer.

4.15. A matrix is invertible if its determinant is nonzero. In other words:

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} a_1 & a_2 \\ a_3 & a_4 \end{vmatrix} \neq 0 \Rightarrow a_1a_4 - a_2a_3 \neq 0$$

Choice (1):

$$\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = 2 \times 5 - 4 \times 3 = -2 \neq 0$$

Choice (2):

$$\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix} = 2 \times 6 - 4 \times 3 = 0$$

Choice (3):

$$\begin{bmatrix} 2 & 8 \\ 3 & 12 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} 2 & 8 \\ 3 & 12 \end{vmatrix} = 2 \times 12 - 8 \times 3 = 0$$

Choice (4):

$$\begin{bmatrix} -2 & -1 \\ 6 & 3 \end{bmatrix} \Rightarrow |A| = \begin{vmatrix} -2 & -1 \\ 6 & 3 \end{vmatrix} = (-2) \times 3 - 6 \times (-1) = 0$$

Choice (1) is the answer.

4.16. The inverse of a matrix can be determined as follows:

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}^{-1} = \frac{1}{a_1a_4 - a_2a_3} \begin{bmatrix} a_4 & -a_2 \\ -a_3 & a_1 \end{bmatrix}$$

Therefore:

$$A = \begin{bmatrix} -1 & b \\ 0 & 1 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} -1 & b \\ 0 & 1 \end{bmatrix}^{-1} = \frac{1}{(-1) \times 1 - 0 \times b} \begin{bmatrix} 1 & -b \\ 0 & -1 \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} -1 & b \\ 0 & 1 \end{bmatrix} \quad (2)$$

Based on the information given in the problem, we have:

$$A^{-1} = \begin{bmatrix} -1 & 3 \\ 0 & 1 \end{bmatrix} \quad (1)$$

Solving (1) and (2):

$$\begin{bmatrix} -1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & b \\ 0 & 1 \end{bmatrix} \Rightarrow b = 3$$

Choice (4) is the answer.

4.17. The system of equations with the form presented in (1) has an infinite number of solutions (the lines match each other) if:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad (1)$$

Hence:

$$\begin{cases} m(x-1) = 3(x-y) \\ 4x + (m+1)y = 2 \end{cases} \Rightarrow \begin{cases} (m-3)x + 3y = m \\ 4x + (m+1)y = 2 \end{cases} \Rightarrow \frac{m-3}{4} = \frac{3}{m+1} = \frac{m}{2} \Rightarrow \begin{cases} \frac{3}{m+1} = \frac{m}{2} \\ \frac{m-3}{4} = \frac{m}{2} \end{cases} \quad (2)$$

$$(2) \Rightarrow \frac{3}{m+1} = \frac{m}{2} \Rightarrow m^2 + m = 6 \Rightarrow m^2 + m - 6 = 0 \Rightarrow (m+3)(m-2) = 0 \Rightarrow m = -3, 2 \quad (3)$$

However, $m = 2$ does not satisfy (3), as is shown in the following:

$$(3) \Rightarrow \frac{m-3}{4} = \frac{m}{2} \Rightarrow \frac{2-3}{4} = \frac{2}{2} \Rightarrow -\frac{1}{4} \neq 1$$

Therefore, only $m = -3$ is acceptable. Choice (2) is the answer.

4.18. The system of equations with the form presented in (1) does not have a solution (the lines are parallel) if:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases} \quad (1)$$

Therefore:

$$\begin{cases} my + 2x = 5 \\ y + (m-1)x = 2m-3 \end{cases} \Rightarrow \begin{cases} 2x + my = 5 \\ (m-1)x + y = 2m-3 \end{cases} \Rightarrow \frac{2}{m-1} = \frac{m}{1} \quad (2)$$

$$\neq \frac{5}{2m-3} \Rightarrow \begin{cases} \frac{2}{m-1} = \frac{m}{1} \\ \frac{m}{1} \neq \frac{5}{2m-3} \end{cases} \quad (3)$$

$$(2) \Rightarrow \frac{2}{m-1} = \frac{m}{1} \Rightarrow m^2 - m = 2 \Rightarrow m^2 - m - 2 = 0 \Rightarrow (m-2)(m+1) = 0 \Rightarrow m = -1, 2$$

However, $m = -1$ does not satisfy (3), as is proven in the following:

$$\frac{m}{1} \neq \frac{5}{2m-3} \Rightarrow \frac{-1}{1} \neq \frac{5}{2 \times (-1) - 3} \Rightarrow -1 \neq -1 \Rightarrow \text{Wrong}$$

Therefore, only $m = 2$ is acceptable. Choice (4) is the answer.

4.19. The problem can be solved as follows:

$$\begin{cases} \frac{3}{2x-1} - \frac{1}{y-2} = \frac{4}{3} \\ \frac{1}{2x-1} + \frac{5}{2-y} = 2 \end{cases} \xrightarrow{\times(-5)} \begin{cases} \frac{-15}{2x-1} + \frac{5}{y-2} = -\frac{20}{3} \\ \frac{1}{2x-1} + \frac{5}{2-y} = 2 \end{cases} \xrightarrow{+} \frac{-15}{2x-1} + \frac{1}{2x-1} = -\frac{20}{3} + 2$$

$$\Rightarrow \frac{-14}{2x-1} = -\frac{14}{3} \Rightarrow 2x-1 = 3 \Rightarrow x = 2$$

$$\frac{1}{2x-1} + \frac{5}{2-y} = 2 \xrightarrow{x=2} \frac{1}{2 \times 2 - 1} + \frac{5}{2-y} = 2 \Rightarrow \frac{5}{2-y} = \frac{5}{3} \Rightarrow 2-y = 3 \Rightarrow y = -1$$

$$\Rightarrow x + y = 2 + (-1) = 1$$

Choice (1) is the answer.

4.20. The system of equations shows the equations of two lines. Therefore, we need to check each pair of equations:

$$\frac{5x+y}{1} = \frac{3x-y}{3} = \frac{7x+y}{2} \Rightarrow \begin{cases} \frac{5x+y}{1} = \frac{3x-y}{3} \\ \frac{3x-y}{3} = \frac{7x+y}{2} \end{cases} \Rightarrow \begin{cases} 15x+3y = 3x-y \\ 6x-2y = 21x+3y \end{cases} \Rightarrow \begin{cases} 12x+4y = 0 \\ 15x+5y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} 3x+y = 0 \\ 3x+y = 0 \end{cases}$$

Therefore, the equations of the lines are the same and the lines match each other. Hence, the system of equations has an infinite number of solutions. Choice (4) is the answer.

Problems: Quadratic Equations

5

Abstract

In this chapter, the basic and advanced problems of quadratic equations are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

- 5.1. Two times of a positive quantity is 9 units fewer than the one-third of its square value. Determine this quantity.

Difficulty level ☒ Easy ☐ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 9
- 2) 12
- 3) 15
- 4) 18

- 5.2. For what value of a , the value of the term of $ax^2 + 2x + 4a$ is always positive?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $a \geq 0.5$
- 2) $a \leq -0.5$
- 3) $0 \leq a \leq 0.5$
- 4) $-0.5 \leq a \leq 0.5$

- 5.3. For what value of k , the roots of the quadratic equation of $2x^2 + 3x - k = 0$ are two units fewer than the quadratic equation of $2x^2 - 5x + 1 = 0$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

- 5.4. For what value of m , the quadratic equation of $2x^2 - 5x + m = 0$ has two real roots that the roots are inverse value of each other?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

5.5. Determine the value of m so that the quadratic equation of $(m + 2)x^2 + 4x + m - 1 = 0$ has two real roots.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $-2 < m < 1$
- 2) $1 < m < 2$
- 3) $-2 < m < 2$
- 4) $-3 < m < 2$

5.6. For what value of m , the inequality of $m^3x^2 + mx + \frac{1}{m} < 0$ is always true?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $|m| < 3$
- 2) $m < 0$
- 3) $m > 0$
- 4) $m \in \mathbb{R}$

5.7. Which one of the quadratic equations below has the roots that their values are nine times of the roots of the quadratic equation of $x^2 + x - 3 = 0$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x^2 + 9x - 243 = 0$
- 2) $x^2 + 9x - 27 = 0$
- 3) $x^2 + 18x - 243 = 0$
- 4) $x^2 + 18x - 27 = 0$

5.8. The sum of the roots of the quadratic equation of $ax^2 + bx + c = 0$ is equal to the product of the reciprocal of the roots. What relation exists between the parameters?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $a^2 + bc = 0$
- 2) $a^2 - bc = 0$
- 3) $b^2 - ac = 0$
- 4) $b^2 + ac = 0$

5.9. For what value of a , the quadratic equation of $2x^2 + ax + a - \frac{3}{2} = 0$ has two distinct roots?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $a < 2$ or $a > 6$
- 2) $a < 3$ or $a > 4$
- 3) $2 < a < 6$
- 4) $3 < a < 4$

5.10. Determine the range of m so that $4x^2 - 2mx + 4m^2 \geq 0$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\mathbb{R}$
- 2) $\{ \}$
- 3) $|m| \leq 2$
- 4) $|m| \geq 2$

- 5.11. In the quadratic equation of $2x^2 + (2k - 1)x - k = 0$, for what value of k , the sum of the inverse value of the roots is equal to $\frac{7}{3}$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -4
 2) -3
 3) 3
 4) 4

- 5.12. Determine the quadratic equation that its roots are $2 + \sqrt{4 - a}$ and $2 - \sqrt{4 - a}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $x^2 - 4x + a = 0$
 2) $x^2 + ax - 4 = 0$
 3) $x^2 + 4x - a = 0$
 4) $x^2 - ax + 4 = 0$

- 5.13. For what value of m , the sum of the square of the roots of $2x^2 - 5x + m = 0$ is $\frac{37}{4}$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -2
 2) -3
 3) 2
 4) 3

- 5.14. If x_1 and x_2 are the roots of the quadratic equation of $mnx^2 + n^2x + m^2 = 0$, determine the value of $x_1^2x_2 + x_1x_2^2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -1
 2) 1
 3) $\frac{m+n}{mn}$
 4) mn

- 5.15. For what value of m , the quadratic equation of $(m + 1)x^2 + m(m^2 - 9)x - 2 = 0$ has two real and symmetric (equal in magnitude, but different in sign) roots.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -1
 2) -3
 3) 3
 4) 9

- 5.16. What is the minimum value of $x^2 - x + 2$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{1}{4}$
 2) $\frac{3}{4}$
 3) $\frac{5}{4}$
 4) $\frac{7}{4}$

- 5.17. Determine the value of k if the relation of $x_1^2 + x_2^2 = 12$ exists between the roots of the quadratic equation of $x^2 - 2kx - 2 = 0$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\pm\sqrt{2}$
 2) $\pm\sqrt{3}$
 3) ± 2
 4) ± 3

- 5.18. The difference of the roots of the quadratic equation of $2x^2 - 3x + m = 0$ is $\frac{5}{2}$. Determine the value of m .

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -2
 2) -1
 3) 1
 4) 2

- 5.19. If the roots of the quadratic equation of $x^2 + ax - a^2 = 0$ are x_1 and x_2 , determine the quadratic equation that its roots are $x_1 + 1$ and $x_2 + 1$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x^2 + (a - 2)x + a^2 + a - 1 = 0$
 2) $x^2 + (a + 2)x - a^2 - a - 1 = 0$
 3) $x^2 + (a - 2)x - a^2 - a + 1 = 0$
 4) $x^2 + (a + 2)x + a^2 + a - 1 = 0$

- 5.20. The sum of the square of two successive numbers is 925. Determine the sum of these two numbers.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 41
 2) 43
 3) 45
 4) 47

- 5.21. If x' and x'' are the roots of the quadratic equation of $x^2 - 4x + 1 = 0$, determine the value of $|\sqrt{x'} - \sqrt{x''}|$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\sqrt{2}$
 2) $\sqrt{3}$
 3) 2
 4) 3

- 5.22. If the quadratic equations of $x^2 - x - 2a = 0$ and $x^2 + 2x + a = 0$ have a common root, determine this common root for $a > 0$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -2
 2) -1
 3) 1

4) 2

5.23. What is the solution of the following inequality?

$$\frac{x-1}{x+1} > 2x$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $x < -1$
- 2) $x > -1$
- 3) $-1 < x < 1$
- 4) $-2 < x < -1$

5.24. How many distinctive real roots do the equation of $(x^2 - 2x)^2 - (x^2 - 2x) = 2$ have?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 1
- 2) 2
- 3) 3
- 4) 4

5.25. For what value of m , the quadratic equation of $mx^2 + 5x + m^2 - 6 = 0$ has two real roots that the roots are inverse value of each other?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $-2, 3$
- 2) -2
- 3) 2
- 4) 3

5.26. For what value of m , the sum of the square of the roots of $2x^2 - mx + m - 1 = 0$ is 4?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $-2, 6$
- 2) -2
- 3) 2
- 4) 6

Abstract

In this chapter, the problems of the fifth chapter are fully solved, in detail, step-by-step, and with different methods.

- 6.1. Based on the information given in the problem, two times of a positive quantity is 9 units fewer than the one-third of its square value. In other words:

$$x > 0 \quad (1)$$

$$2x = \frac{1}{3}x^2 - 9 \quad (2)$$

$$\Rightarrow x^2 - 6x - 27 = 0 \Rightarrow (x - 9)(x + 3) = 0 \Rightarrow x = -3, 9 \quad (3)$$

Solving (1) and (3):

$$x = 9$$

Choice (1) is the answer.

- 6.2. The value of the quadratic equation of $ax^2 + bx + c$ is always positive if $a > 0$ and $\Delta \leq 0$.

Therefore, for the given quadratic equation of $ax^2 + 2x + 4a$, we can write:

$$a > 0 \quad (1)$$

$$\Delta = 2^2 - 4(a)(4a) \leq 0 \Rightarrow 4 - 16a^2 \leq 0 \Rightarrow a^2 \geq \frac{1}{4} \Rightarrow a \geq \frac{1}{2} \& a \leq -\frac{1}{2} \quad (2)$$

Solving (1) and (2):

$$a \geq \frac{1}{2}$$

Choice (1) is the answer.

- 6.3. Based on the information given in the problem, we have:

$$2x^2 + 3x - k = 0 \quad (1)$$

$$2X^2 - 5X + 1 = 0 \quad (2)$$

$$x = X - 2 \Rightarrow X = x + 2 \quad (3)$$

Solving (2) and (3):

$$2(x+2)^2 - 5(x+2) + 1 = 0 \Rightarrow 2x^2 + 3x - 1 = 0 \quad (4)$$

Comparing (1) and (4):

$$k = 1$$

Choice (1) is the answer.

6.4. The product of the roots of a quadratic equation can be determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow \text{Product of roots} = \frac{c}{a}$$

Since the roots are inverse value of each other, the product of the roots is equal to one. Hence:

$$2x^2 - 5x + m = 0 \Rightarrow \text{Product of roots} = \frac{m}{2} = 1 \Rightarrow m = 2$$

Choice (2) is the answer.

6.5. A quadratic equation has two real roots if its discriminant is equal or greater than zero. In other words:

$$ax^2 + bx + c = 0 \Rightarrow \Delta = b^2 - 4ac \geq 0$$

Therefore, for the quadratic equation of $(m+2)x^2 + 4x + m - 1 = 0$, we can write:

$$\Delta = 4^2 - 4(m+2)(m-1) \geq 0 \Rightarrow 16 - 4(m^2 + m - 2) \geq 0 \Rightarrow m^2 + m - 6 \leq 0$$

$$\Rightarrow (m+3)(m-2) \leq 0 \Rightarrow -3 \leq m \leq 2$$

Choice (4) is the answer.

6.6. The value of the quadratic equation of $ax^2 + bx + c$ is always negative if $a < 0$ and $\Delta < 0$.

Therefore, for the given quadratic equation of $m^3x^2 + mx + \frac{1}{m}$, we can write:

$$m^3 < 0 \Rightarrow m < 0 \quad (1)$$

$$\Delta = m^2 - 4(m^3)\left(\frac{1}{m}\right) < 0 \Rightarrow m^2 - 4m^2 < 0 \Rightarrow -3m^2 < 0 \Rightarrow m^2 > 0 \Rightarrow m \in \mathbb{R} \quad (2)$$

Solving (1) and (2):

$$m < 0$$

Choice (2) is the answer.

6.7. Based on the information given in the problem, we know that:

$$X = 9x \Rightarrow x = \frac{X}{9} \quad (1)$$

$$x^2 + x - 3 = 0 \quad (2)$$

Solving (1) and (2):

$$\left(\frac{X}{9}\right)^2 + \frac{X}{9} - 3 = 0 \Rightarrow X^2 + 9X - 243 = 0$$

Choice (1) is the answer.

6.8. We know that a quadratic equation has the form below, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (1)$$

Based on the information given in the problem, we know that:

$$x_1 + x_2 = \frac{1}{x_1} \times \frac{1}{x_2} = \frac{1}{x_1 x_2} \quad (2)$$

Solving (1) and (2):

$$-\frac{b}{a} = \frac{1}{\frac{c}{a}} \Rightarrow -\frac{b}{a} = \frac{a}{c} \Rightarrow a^2 = -bc \Rightarrow a^2 + bc = 0$$

Choice (1) is the answer.

6.9. A quadratic equation has two real distinct roots if its discriminant is greater than zero. In other words:

$$ax^2 + bx + c = 0 \Rightarrow \Delta = b^2 - 4ac > 0$$

Therefore:

$$2x^2 + ax + a - \frac{3}{2} = 0 \Rightarrow \Delta = a^2 - 4 \times 2 \times \left(a - \frac{3}{2}\right) > 0 \Rightarrow a^2 - 8a + 12 > 0$$

$$\Rightarrow (a - 6)(a - 2) > 0 \Rightarrow a < 2 \text{ \& } a > 6$$

Choice (1) is the answer.

6.10. The value of the quadratic equation of $ax^2 + bx + c$ is always positive if $a > 0$ and $\Delta \leq 0$.

Therefore, for the given quadratic equation of $4x^2 - 2mx + 4m^2 \geq 0$, we can write:

$$4 > 0 \quad (1)$$

$$\Delta = (-2m)^2 - 4(4)(4m^2) \leq 0 \Rightarrow 4m^2 - 64m^2 \leq 0 \Rightarrow -60m^2 \leq 0 \Rightarrow m^2 \geq 0 \quad (2)$$

As can be seen in (1) and (2), both criteria are satisfied disregarding the value of m . Therefore:

$$m \in \mathbb{R}$$

Choice (1) is the answer.

6.11. Based on the information given in the problem, we have:

$$\frac{1}{x_1} + \frac{1}{x_2} = \frac{7}{3} \Rightarrow \frac{x_1 + x_2}{x_1 x_2} = \frac{7}{3} \quad (1)$$

As we know, the sum and the product of the roots of a quadratic equation can be determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a}$$

Hence, for the quadratic equation of $2x^2 + (2k - 1)x - k = 0$, we can write:

$$x_1 + x_2 = -\frac{(2k - 1)}{2} \quad (2)$$

$$x_1 x_2 = \frac{-k}{2} \quad (3)$$

Solving (1), (2), and (3):

$$\frac{-\frac{(2k-1)}{2}}{\frac{-k}{2}} = \frac{7}{3} \Rightarrow \frac{2k-1}{k} = \frac{7}{3} \Rightarrow 6k-3=7k \Rightarrow k = -3$$

Choice (2) is the answer.

6.12. A quadratic equation has the following form, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots}, \quad P = \text{product of roots} \quad (1)$$

Therefore, for the given roots of $2 + \sqrt{4 - a}$ and $2 - \sqrt{4 - a}$, we can write:

$$S = (2 + \sqrt{4 - a}) + (2 - \sqrt{4 - a}) = 4 \quad (2)$$

$$P = (2 + \sqrt{4 - a})(2 - \sqrt{4 - a}) = 4 - (4 - a) = a \quad (3)$$

Solving (1), (2), and (3):

$$x^2 - 4x + a = 0$$

Choice (1) is the answer.

6.13. Based on the information given in the problem, we know that:

$$x_1^2 + x_2^2 = \frac{37}{4} \Rightarrow (x_1 + x_2)^2 - 2x_1 x_2 = \frac{37}{4} \quad (1)$$

On the other hand, we know that a quadratic equation has the form below, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (2)$$

Solving (1) and (2):

$$S^2 - 2P = \frac{37}{4} \Rightarrow \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{37}{4} \quad (3)$$

Solving (3) for the given quadratic equation of $2x^2 - 5x + m = 0$:

$$\left(-\frac{-5}{2}\right)^2 - 2\frac{m}{2} = \frac{37}{4} \Rightarrow \frac{25}{4} - m = \frac{37}{4} \Rightarrow m = -3$$

Choice (2) is the answer.

6.14. The sum and the product of the roots of a quadratic equation can be determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (1)$$

The problem can be solved as follows:

$$x_1^2 x_2 + x_1 x_2^2 = x_1 x_2 (x_1 + x_2) \quad (2)$$

Solving (1) and (2):

$$x_1^2 x_2 + x_1 x_2^2 = P \times S = \left(-\frac{b}{a}\right) \left(\frac{c}{a}\right) \quad (3)$$

For the quadratic equation of $mnx^2 + n^2x + m^2 = 0$, we can write:

$$x_1^2 x_2 + x_1 x_2^2 = \left(-\frac{n^2}{mn}\right) \left(\frac{m^2}{mn}\right) = -1$$

Choice (1) is the answer.

6.15. Based on the information given in the problem, we have:

$$x_1 = -x_2 \Rightarrow x_1 + x_2 = 0 \quad (1)$$

As we know, the sum and the product of the roots of a quadratic equation are determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (2)$$

Solving (1) and (2):

$$S = -\frac{b}{a} = 0 \Rightarrow b = 0 \quad (3)$$

Hence, for the given quadratic equation of $(m+1)x^2 + m(m^2-9)x - 2 = 0$, we can write:

$$m(m^2-9) = 0 \Rightarrow m = 0, -3, 3 \quad (4)$$

On the other hand, to have real roots for the given quadratic equation, the constraint below must be satisfied:

$$\Delta \geq 0 \Rightarrow m^2(m^2-9)^2 + 8(m+1) \geq 0 \Rightarrow 8(m+1) \geq 0 \Rightarrow m \geq -1 \quad (5)$$

Solving (4) and (5):

$$\Rightarrow m = 0, 3$$

However, $m = 0$ does not exist in the choices. Therefore, $m = 3$ is the only acceptable choice. Choice (3) is the answer.

6.16. The problem can be solved as follows:

$$x^2 - x + 2 = x^2 - x + \frac{1}{4} + \frac{7}{4} = x^2 - 2\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)^2 + \frac{7}{4} = \left(x - \frac{1}{2}\right)^2 + \frac{7}{4}$$

The term of $\left(x - \frac{1}{2}\right)^2$ is equal or greater than zero. Hence:

$$\left(x - \frac{1}{2}\right)^2 \geq 0 \Rightarrow \left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \geq \frac{7}{4} \Rightarrow \min \left\{ \left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \right\} = \frac{7}{4}$$

Choice (4) is the answer.

6.17. Based on the information given in the problem, we know that:

$$x_1^2 + x_2^2 = 12 \Rightarrow (x_1 + x_2)^2 - 2x_1x_2 = 12 \quad (1)$$

On the other hand, we know that a quadratic equation has the form below, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (2)$$

Solving (1) and (2):

$$S^2 - 2P = 12 \Rightarrow \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = 12 \quad (3)$$

Solving (3) for the given quadratic equation of $x^2 - 2kx - 2 = 0$:

$$\left(-\frac{-2k}{1}\right)^2 - 2\left(\frac{-2}{1}\right) = 12 \Rightarrow 4k^2 + 4 = 12 \Rightarrow k^2 = 2 \Rightarrow k = \pm\sqrt{2}$$

Choice (1) is the answer.

6.18. We know that a quadratic equation has the form below, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a}$$

Therefore, for the given quadratic equation of $2x^2 - 3x + m = 0$, we can write:

$$x_1 + x_2 = S = -\frac{-3}{2} = \frac{3}{2}$$

Based on the information given in the problem, we have:

$$x_1 - x_2 = \frac{5}{2}$$

Hence:

$$\Rightarrow \begin{cases} x_1 + x_2 = \frac{3}{2} \\ x_1 - x_2 = \frac{5}{2} \end{cases} \Rightarrow 2x_1 = 4 \Rightarrow x_1 = 2$$

$$x_1 + x_2 = \frac{3}{2} \xrightarrow{x_1 = 2} x_2 = -\frac{1}{2}$$

$$x_1x_2 = P = \frac{c}{a} \Rightarrow 2 \times \left(-\frac{1}{2}\right) = \frac{m}{2} \Rightarrow m = -2$$

Choice (1) is the answer.

6.19. Based on the information given in the problem, x_1 and x_2 are the roots of the quadratic equation of $x^2 + ax - a^2 = 0$.

To determine the quadratic equation that its roots are $y_1 = x_1 + 1$ and $y_2 = x_2 + 1$, we need to replace x by $y - 1$ in the same quadratic equation, as follows:

$$(y - 1)^2 + a(y - 1) - a^2 = 0 \Rightarrow y^2 - 2y + 1 + ay - a - a^2 = 0$$

$$\Rightarrow y^2 + (a-2)y + 1 - a - a^2 = 0 \xrightarrow{y \rightarrow x} x^2 + (a-2)x + 1 - a - a^2 = 0$$

Choice (3) is the answer.

- 6.20. Based on the information given in the problem, we know that the sum of the square of the two successive numbers (e.g., n and $n+1$) is 925. In other words:

$$n^2 + (n+1)^2 = 925 \Rightarrow n^2 + n^2 + 2n + 1 = 925 \Rightarrow 2n^2 + 2n = 924 \Rightarrow n(n+1) = 462 \quad (1)$$

We can consider 462 as follows:

$$462 = 21 \times 22 \quad (2)$$

Solving (1) and (2):

$$n(n+1) = 21 \times 22 \Rightarrow n = 21$$

$$\Rightarrow \text{The sum of the two numbers} = n + (n+1) = 21 + 22 = 43$$

Choice (2) is the answer.

- 6.21. The sum and the product of the roots of a quadratic equation can be determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (1)$$

The problem can be solved as follows:

$$\left(\sqrt{x'} - \sqrt{x''}\right)^2 = x' + x'' - 2\sqrt{x'x''} \quad (2)$$

$$x^2 - 4x + 1 = 0 \Rightarrow S = -\frac{b}{a} = -\frac{-4}{1} = 4, P = \frac{c}{a} = \frac{1}{1} = 1 \quad (3)$$

Solving (1), (2), and (3):

$$\left(\sqrt{x'} - \sqrt{x''}\right)^2 = x' + x'' - 2\sqrt{x'x''} = S - 2\sqrt{P} = 4 - 2\sqrt{1} = 2 \Rightarrow \left|\sqrt{x'} - \sqrt{x''}\right| = \sqrt{2}$$

Choice (1) is the answer.

- 6.22. Based on the information given in the problem, we have:

$$a > 0 \quad (1)$$

If it is assumed that x_1 is the common root, then we can write:

$$\begin{cases} x_1^2 + 2x_1 + a = 0 \\ x_1^2 - x_1 - 2a = 0 \end{cases} \Rightarrow 3x_1 + 3a = 0 \Rightarrow x_1 = -a \quad (2)$$

By replacing $x_1 = -a$ in one of the equations, we have:

$$x_1^2 + 2x_1 + a = 0 \xrightarrow{x_1 = -a} (-a)^2 + 2(-a) + a = 0 \Rightarrow a^2 - a = 0 \Rightarrow a(a-1) = 0 \Rightarrow a = 0, 1 \quad (3)$$

Solving (1) and (3):

$$a = 1 \xrightarrow{(2)} x_1 = -1$$

Choice (2) is the answer.

6.23. The problem can be solved as follows:

$$\frac{x-1}{x+1} > 2x \Rightarrow \frac{x-1}{x+1} - 2x > 0 \Rightarrow \frac{x-1-2x^2-2x}{x+1} > 0 \Rightarrow \frac{2x^2+x+1}{x+1} < 0$$

The term of $2x^2 + x + 1$ is always positive because $a > 0$ and $\Delta < 0$, as can be seen in the following:

$$a = 2 > 0$$

$$\Delta = 1 - 4 \times 2 \times 1 = -7 < 0$$

Therefore:

$$x + 1 < 0 \Rightarrow x < -1$$

Choice (1) is the answer.

6.24. **First Method:** By defining the new variable below, the equation of $(x^2 - 2x)^2 - (x^2 - 2x) = 2$ can be simplified and changed to a quadratic equation as follows:

$$X \equiv x^2 - 2x$$

$$\Rightarrow X^2 - X - 2 = 0$$

$$\Rightarrow X_1, X_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times (-2)}}{2 \times 1} = \frac{1 \pm \sqrt{9}}{2} = \frac{1 \pm 3}{2}$$

$$\Rightarrow X_1 = \frac{1+3}{2}, X_2 = \frac{1-3}{2} \Rightarrow X_1 = 2, X_2 = -1$$

Now, we need to come back to the primary variable (x). Therefore, we have two separate quadratic equations:

$$x^2 - 2x = 2$$

$$x^2 - 2x = -1$$

To solve $x^2 - 2x - 2 = 0$:

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times (-2)}}{2 \times 1} = \frac{2 \pm \sqrt{12}}{2} = 1 \pm \sqrt{3}$$

$$\Rightarrow x_1 = 1 + \sqrt{3}, x_2 = 1 - \sqrt{3}$$

As can be seen, this equation includes two distinctive real roots.

To solve $x^2 - 2x + 1 = 0$, we do not need to apply the quadratic equation's formula, since it is a binomial expression:

$$(x-1)^2 = 0 \Rightarrow (x-1) = 0 \Rightarrow x = 1 \text{ (Double root)} \Rightarrow x_3 = 1, x_4 = 1$$

As can be seen, this equation has only one distinctive real root.

Therefore, the main equation has three distinctive real roots, in overall. **Choice (3) is the answer.**

Second Method: In this method, we do not need to solve the quadratic equations. In fact, we can determine the number of distinctive real roots based on the value of discriminant of each quadratic equation ($\Delta = b^2 - 4ac$).

Regarding the equation of $x^2 - 2x - 2 = 0$:

$$\Delta = b^2 - 4ac = (-2)^2 - 4 \times 1 \times (-2) = 12 > 0$$

Therefore, we will have two distinctive real roots, since the discriminant is greater than zero.

Moreover, regarding the equation of $x^2 - 2x + 1 = 0$:

$$\Delta = b^2 - 4ac = (-2)^2 - 4 \times 1 \times 1 = 0$$

Thus, we will have just one distinctive real root, as the discriminant is equal to zero.

In overall, we will have three distinctive real roots. **Choice (3) is the answer.**

6.25. A quadratic equation has two real distinct roots if its discriminant is greater than zero. In other words:

$$ax^2 + bx + c = 0 \Rightarrow \Delta = b^2 - 4ac > 0$$

Therefore:

$$mx^2 + 5x + m^2 - 6 = 0 \Rightarrow \Delta = 5^2 - 4m(m^2 - 6) > 0 \quad (1)$$

Moreover, the product of the roots of a quadratic equation can be determined as follows:

$$ax^2 + bx + c = 0 \Rightarrow \text{Product of roots} = \frac{c}{a}$$

Since the roots are inverse value of each other, the product of the roots is equal to one. Hence:

$$\begin{aligned} mx^2 + 5x + m^2 - 6 = 0 \Rightarrow \text{Product of roots} &= \frac{m^2 - 6}{m} = 1 \Rightarrow m^2 - m - 6 = 0 \\ \Rightarrow (m+2)(m-3) &= 0 \Rightarrow m = -2, 3 \end{aligned} \quad (2)$$

Solving (1) and (2) for $m = 3$:

$$\Delta = 5^2 - 4m(m^2 - 6) \xrightarrow{m=3} 5^2 - 4 \times 3(3^2 - 6) = 25 - 12 \times 3 = -11 < 0$$

Therefore, $m = 3$ is not acceptable.

Solving (1) and (2) for $m = -2$:

$$\Delta = 5^2 - 4m(m^2 - 6) \xrightarrow{m=-2} 5^2 - 4 \times (-2)((-2)^2 - 6) = 25 - 16 = 9 > 0$$

Thus, $m = -2$ is not acceptable. **Choice (2) is the answer.**

6.26. Based on the information given in the problem, we know that:

$$x_1^2 + x_2^2 = 4 \Rightarrow (x_1 + x_2)^2 - 2x_1x_2 = 4 \quad (1)$$

On the other hand, we know that a quadratic equation has the form below, where S and P are the sum and the product of its roots:

$$x^2 - Sx + P = 0, \quad S = \text{sum of roots} = -\frac{b}{a}, \quad P = \text{product of roots} = \frac{c}{a} \quad (2)$$

Solving (1) and (2):

$$S^2 - 2P = 4 \Rightarrow \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = 4 \quad (3)$$

Solving (3) for the given quadratic equation of $2x^2 - mx + m - 1 = 0$:

$$\begin{aligned} \left(-\frac{-m}{2}\right)^2 - 2\frac{m-1}{2} &= 4 \Rightarrow \frac{m^2}{4} - m + 1 = 4 \Rightarrow \frac{m^2}{4} - m - 3 = 0 \Rightarrow m^2 - 4m - 12 = 0 \\ &\Rightarrow (m-6)(m+2) = 0 \Rightarrow m = 6, -2 \end{aligned}$$

$m = 6$ is not acceptable because the equation will not have real roots, as can be seen in the following:

$$2x^2 - mx + m - 1 = 0 \xrightarrow{m=6} 2x^2 - 6x + 5 = 0 \Rightarrow \Delta = (-6)^2 - 4(2)(5) = 36 - 40 = -4 < 0$$

$m = -2$ is acceptable because the equation will have two real roots, as is presented below:

$$2x^2 - mx + m - 1 = 0 \xrightarrow{m=-2} 2x^2 + 2x - 3 = 0 \Rightarrow \Delta = (2)^2 - 4(2)(-3) = 4 + 24 = 28 > 0$$

Choice (2) is the answer.

Problems: Functions, Algebra of Functions, and Inverse Functions

7

Abstract

In this chapter, the basic and advanced problems of functions, algebra of functions, and inverse functions are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

7.1. If $f\left(\frac{1}{x}\right) = \sqrt{\frac{2x-1}{x^2}}$ and $g(x) = 2\cos^2(x)$, calculate the value of $fog\left(\frac{\pi}{3}\right)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 0
- 2) $\frac{1}{2}$
- 3) $\frac{\sqrt{3}}{2}$
- 4) 2

7.2. Determine the value of $fog(3)$ if $f(x) = \sqrt{x} - 2$ and $g(x) = x + 1$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 0
- 2) 1
- 3) 2
- 4) 3

7.3. If $f(x) = 2x - 2$ and $g(x) = x^2 - 1$, solve the equation of $fog(x) = 0$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\pm\sqrt{2}$
- 2) ± 2
- 3) $\pm\sqrt{3}$
- 4) ± 3

7.4. In the function below, calculate the value of $f(-2) + f(2)$:

$$f(x) = \begin{cases} 2x^2 + 4 & x \geq 2 \\ [x] - 4 & x < 2 \end{cases}$$

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 8
- 2) 6
- 3) 10
- 4) 5

7.5. Determine the domain of $y = \sqrt{2 - x^2}$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x \leq -\sqrt{2}, x \geq \sqrt{2}$
- 2) $-\sqrt{2} \leq x \leq \sqrt{2}$
- 3) $x = 0$
- 4) $-\sqrt{2} < x < \sqrt{2}$

7.6. Calculate $f \circ g(x)$ if $f(x) = 1 - x^2$ and $g(x) = \sin(x)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\cos^2(x)$
- 2) $\cos(x)$
- 3) $\sin(1 - x^2)$
- 4) $\sin(\cos(x))$

7.7. What is the inverse function of $f(x) = \frac{1}{x}$?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) x
- 2) $\frac{1}{x}$
- 3) $-\frac{1}{x}$
- 4) $\sqrt{x}$

7.8. Calculate $f \circ f(x)$ if $f(x) = \frac{1-x}{1+x}$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ○ Small ● Normal ○ Large

- 1) $\left(\frac{1+x}{1-x}\right)^2$
- 2) 1
- 3) x
- 4) $\left(\frac{1-x}{1+x}\right)^2$

7.9. Determine the inverse function of $f(x) = x^2 - 2x$

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $1 + \sqrt{x+1}$
- 2) $1 - \sqrt{x+1}$
- 3) $1 + \sqrt{x-1}$
- 4) $1 - \sqrt{x-1}$

7.10. Determine the value of $f(x)$ if $f(x+1) = x^2 - 2x + 1$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $(x-2)^2$
- 2) $(x-1)^2$
- 3) $x^2 - 2x$
- 4) $(x+2)^2$

7.11. Which one of the terms below is not a function?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $y^2 = x$
- 2) $y^3 = x$
- 3) $y = \sqrt{x^2}$
- 4) $y = \begin{cases} x^2 & x \geq 0 \\ 1 & x < 0 \end{cases}$

7.12. If $f(\sqrt{x}) = x + \sqrt{x}$, calculate the value of $f(2) + f(1)$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 6
- 2) 7
- 3) 8
- 4) 9

7.13. Calculate the value of $f(f(0))$ if:

$$f(x) = \begin{cases} x^2 + 1 & x \geq 1 \\ 2x + 3 & x < 1 \end{cases}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 3
- 2) 5
- 3) 10
- 4) 26

7.14. Determine the domain of the function below:

$$f(x) = \sqrt{\frac{1-|x|}{1+|x|}}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\mathbb{R}$
- 2) $x \leq 1$
- 3) $x \geq 1$
- 4) $-1 \leq x \leq 1$

7.15. Determine the domain of the function below:

$$f(x) = \sqrt{\frac{x-1}{x-3}} + \sqrt{\frac{2-x}{x}}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $(0, 1]$
- 2) $[0, 1]$
- 3) $(0, 2]$
- 4) $(1, 3)$

7.16. Determine the domain of the function below:

$$f(x) = \frac{\sqrt{x}}{|x| - 1}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\mathbb{R}$
- 2) $[0, \infty) - \{1\}$
- 3) $\mathbb{R} - \{1\}$
- 4) $[0, \infty)$

7.17. Determine the domain of the function below:

$$f(x) = \frac{\sqrt{x} + 1}{x\sqrt{x} + 1}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\mathbb{R} - \{0\}$
- 2) $[1, \infty)$
- 3) $[0, \infty)$
- 4) $\mathbb{R} - \{-1, 0\}$

7.18. Which number does not exist in the domain of the function below?

$$f(x) = \frac{1-x}{4x+x^3}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -2
- 2) 1
- 3) 2
- 4) 0

7.19. Determine $g(x)$ if $f(x) = 2x$ and $f(g(x)) = 2x + 2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x - 1$
- 2) $x + 2$
- 3) $x + 1$
- 4) $x - 2$

7.20. Determine $g(x)$ if $f(x) = x - 1$ and $f(g(x)) = x$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x + 1$
- 2) $x^2 - x$
- 3) $2x - 1$
- 4) $x^2 - 1$

7.21. Calculate the inverse function of $f(x) = \sqrt{1-x}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $f^{-1}(x) = 1 - x^2, x \geq 0$
- 2) $f^{-1}(x) = \frac{1}{\sqrt{1-x}}$
- 3) $f^{-1}(x) = \sqrt{1+x}$
- 4) $f^{-1}(x) = 1 - x^2$

7.22. What is the inverse function of $f(x) = \sin(x) - 2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $2\arcsin(x)$
- 2) $-2\arcsin(x)$
- 3) $\arcsin(x - 2)$
- 4) $\arcsin(x + 2)$

7.23. Calculate the inverse function of $f \circ g(x)$ if $f(x) = 3x - 2$ and $g(x) = 2 + x$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\frac{1}{3}x - \frac{4}{3}$
- 2) $3x - 4$
- 3) $\frac{1}{3}x + \frac{4}{3}$
- 4) $3x + 4$

7.24. Calculate the inverse function of $f(x) = x^3 + 3x^2 + 3x + 2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $1 - \sqrt[3]{x-1}$
- 2) $1 - \sqrt[3]{x+1}$
- 3) $-1 + \sqrt[3]{x-1}$
- 4) $-1 - \sqrt[3]{x+1}$

7.25. If $f(x) + xf(-x) = x^2 + 1$, then what is the value of $f(2)$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -1
- 2) -2
- 3) 3
- 4) 4

7.26. Calculate the value of $f(f(f(2 \cos(x))))$ if $f(x) = x^2 - 2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $2\sin^8(x)$
- 2) $2\cos^8(x)$
- 3) $2\sin(8x)$
- 4) $2\cos(8x)$

7.27. Determine the domain of the following function:

$$f\left(\frac{x-1}{x}\right) = \sqrt{2x-1}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $[-1, 0)$
- 2) $[-1, 1]$
- 3) $[-1, 1)$
- 4) $[1, \infty)$

7.28. Determine the domain of $f(x) = \sqrt{1 - \sqrt{x-1}}$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $x \geq 1$
- 2) $1 \leq x \leq 2$
- 3) $x \leq 2$
- 4) $\frac{5}{4} \leq x \leq \frac{7}{4}$

7.29. Determine the domain of the function of $f(x) = \sqrt{|x|-1} + \sqrt{|x|+1}$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\mathbb{R} - [-1, 1]$
- 2) $\mathbb{R}$
- 3) $[-1, 1]$
- 4) $\mathbb{R} - (-1, 1)$

7.30. Calculate the value of $f(3)$ if:

$$f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{28}{3}$
- 2) $\frac{1}{7}$
- 3) 7
- 4) $\frac{3}{28}$

7.31. For what value of a , the function of $f(x) = |x+2| + a|x-2|$ is even?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -1
- 2) 0
- 3) 1
- 4) 2

7.32. The functions of $f(x) = x^2 + (A - 1)x$ and $g(x) = (B + 2)x^2 + \sin(x)$ are even and odd, respectively. Calculate the value of $A + B$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) -2
- 2) -1
- 3) 1
- 4) 2

7.33. Which one of the following functions is odd?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $\arccos(\cos(x))$
- 2) $\sqrt{1-x} - \sqrt{1+x}$
- 3) $x^4 + x$
- 4) $x \sin(x)$

7.34. Which one of the functions below is odd?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $|x - 1| + |x + 1|$
- 2) $\sin(|x|)$
- 3) $x^3 + x^2$
- 4) $|x - 1| - |x + 1|$

7.35. Which one of the functions below is even?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $|x - 1| + |x + 1| + |x|$
- 2) $(x + 1)^4$
- 3) $f^2(x) + \sqrt[3]{x-1} = 0$
- 4) $f(x) = [x] + 1$

7.36. We know that $f(g(x)) = x^2 + \frac{1}{x^2} - 4$ and $g(x) = x - \frac{1}{x}$. Determine $f(x)$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x^2 - 4$
- 2) $x^2 - 2$
- 3) x^2
- 4) $x^2 + 2$

7.37. Calculate the value of $f(-f(x))$ if $f(x) = \begin{cases} x^2 + 1 & x > 0 \\ 1 & x \leq 0 \end{cases}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) 1
- 2) $x + 1$
- 3) $x^2 + 1$
- 4) $(x^2 + 1)^2 + 1$

7.38. Determine the domain of the function below:

$$f(x) = \sqrt{\log\left(\frac{5x-x^2}{4}\right)}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $1 < x < 4$
- 2) $0 < x < 5$
- 3) $1 \leq x \leq 4$
- 4) $0 \leq x \leq 5$

7.39. Determine the domain of $f(x) = \sqrt{\log_x(x^2 + 9)}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $(-\infty, \infty)$
- 2) $(0, \infty)$
- 3) $[-3, 3]$
- 4) $x > 0 - \{1\}$

7.40. Calculate the range of the function of $f(x) = 2x - 2[x] + 1$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $[0, 2]$
- 2) $[1, 3)$
- 3) $[0, 2)$
- 4) $[0, 3]$

7.41. Calculate the range of $f \circ g(x)$ if $f(x) = x^2 + 1$ and $g(x) = \sqrt{x-1}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $[0, \infty)$
- 2) $[1, \infty)$
- 3) $[-1, \infty)$
- 4) $\mathbb{R}$

7.42. Calculate the range of $f(x) = \sqrt{x^2 - 2x + 3}$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $[\sqrt{2}, \infty)$
- 2) $[\sqrt{3}, \infty)$
- 3) $[0, \infty)$
- 4) $[1, \infty)$

7.43. Which one of the following functions is equivalent to $f(x) = |x - 2|$?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☐ Normal ☒ Large

- 1) $g_1(x) = \left| \frac{x^2 - 3x + 2}{x - 1} \right|$
- 2) $g_2(x) = \left| \frac{x^2 - 4}{x + 2} \right|$
- 3) $g_3(x) = \frac{(x-2)^2}{|x-2|}$
- 4) $g_4(x) = \frac{|6x-12|}{6}$

Solutions of Problems: Functions, Algebra of Functions, and Inverse Functions

8

Abstract

In this chapter, the problems of the seventh chapter are fully solved, in detail, step-by-step, and with different methods.

8.1. Based on the information given in the problem, we have:

$$f\left(\frac{1}{x}\right) = \sqrt{\frac{2x-1}{x^2}}$$

$$g(x) = 2 \cos^2(x)$$

The problem can be solved as follows:

$$\Rightarrow fog\left(\frac{\pi}{3}\right) = f\left(g\left(\frac{\pi}{3}\right)\right) = f\left(2 \cos^2\left(\frac{\pi}{3}\right)\right) = f\left(2\left(\frac{1}{2}\right)^2\right) = f\left(\frac{1}{2}\right) = \sqrt{\frac{2(2)-1}{2^2}} = \frac{\sqrt{3}}{2}$$

Choice (3) is the answer.

8.2. Based on the information given in the problem, we have:

$$f(x) = \sqrt{x} - 2$$

$$g(x) = x + 1$$

The problem can be solved as follows:

$$\Rightarrow fog(3) = f(g(3)) = f(3 + 1) = f(4) = \sqrt{4} - 2 = 0$$

Choice (1) is the answer.

8.3. Based on the information given in the problem, we have:

$$f(x) = 2x - 2$$

$$g(x) = x^2 - 1$$

The problem can be solved as follows:

$$\Rightarrow fog(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) - 2 = 2x^2 - 4$$

$$fog(x) = 0 \Rightarrow 2x^2 - 4 = 0 \Rightarrow x^2 = 2 \Rightarrow x = \pm \sqrt{2}$$

Choice (1) is the answer.

8.4. Based on the information given in the problem, we have:

$$f(x) = \begin{cases} 2x^2 + 4 & x \geq 2 \\ [x] - 4 & x < 2 \end{cases}$$

The problem can be solved as follows:

$$\Rightarrow f(2) = 2(2)^2 + 4 = 12$$

$$\Rightarrow f(-2) = [-2] - 4 = -6$$

$$\Rightarrow f(2) + f(-2) = 12 + (-6) = 6$$

Choice (2) is the answer.

8.5. Based on the information given in the problem, we have:

$$y = \sqrt{2 - x^2}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Therefore:

$$2 - x^2 \geq 0 \Rightarrow x^2 \leq 2 \Rightarrow -\sqrt{2} \leq x \leq \sqrt{2}$$

Choice (2) is the answer.

8.6. From trigonometry, we know that:

$$\sin^2(x) + \cos^2(x) = 1$$

Based on the information given in the problem, we have:

$$f(x) = 1 - x^2$$

$$g(x) = \sin(x)$$

Therefore:

$$fog(x) = f(g(x)) = f(\sin(x)) = 1 - (\sin(x))^2 = \cos^2(x)$$

Choice (1) is the answer.

8.7. Based on the information given in the problem, we have:

$$f(x) = \frac{1}{x}$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$y = \frac{1}{x} \Rightarrow x = \frac{1}{y} \Rightarrow y = \frac{1}{x} \stackrel{\text{or}}{\Rightarrow} f^{-1}(x) = \frac{1}{x}$$

Choice (2) is the answer.

8.8. Based on the information given in the problem, we have:

$$f(x) = \frac{1-x}{1+x}$$

Therefore:

$$f \circ f(x) = f(f(x)) = f\left(\frac{1-x}{1+x}\right) = \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}} = \frac{\frac{1+x-(1-x)}{1+x}}{\frac{1+x+1-x}{1+x}} = \frac{\frac{2x}{1+x}}{\frac{2}{1+x}} = x$$

Choice (3) is the answer.

8.9. Based on the information given in the problem, we have:

$$f(x) = x^2 - 2x$$

First, we need to define the function in a square form as follows:

$$\Rightarrow f(x) = x^2 - 2x + 1 - 1 \Rightarrow f(x) = (x-1)^2 - 1$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$\Rightarrow y = (x-1)^2 - 1 \Rightarrow y+1 = (x-1)^2 \Rightarrow \sqrt{y+1} = x-1 \Rightarrow \sqrt{y+1} + 1 = x \Rightarrow y = \sqrt{y+1} + 1$$

$$\stackrel{\text{or}}{\Rightarrow} f^{-1}(x) = 1 + \sqrt{y+1}$$

Choice (1) is the answer.

8.10. Based on the information given in the problem, we have:

$$f(x+1) = x^2 - 2x + 1$$

The problem can be solved as follows:

$$\underline{\underline{x \rightarrow x-1}} \Rightarrow f((x-1)+1) = (x-1)^2 - 2(x-1) + 1 \Rightarrow f(x) = x^2 - 2x + 1 - 2x + 2 + 1$$

$$\Rightarrow f(x) = x^2 - 4x + 4 \Rightarrow f(x) = (x-2)^2$$

Choice (1) is the answer.

8.11. A mathematical relation is a function if for any value of x , one value of y is achieved at most. Or, a function is a binary relation between two sets that associates every element of the first set to exactly one element of the second set. Herein, $y^2 = x$ is not a function because for $x = 1$, $y = -1, 1$ are achieved. **Choice (1) is the answer.**

8.12. Based on the information given in the problem, we have:

$$f(\sqrt{x}) = x + \sqrt{x}$$

The problem can be solved as follows:

$$\begin{aligned} f(\sqrt{x}) = x + \sqrt{x} &= (\sqrt{x})^2 + \sqrt{x} \xrightarrow{\sqrt{x} \rightarrow x} f(x) = x^2 + x \\ \Rightarrow f(2) + f(1) &= 2^2 + 2 + 1^2 + 1 = 8 \end{aligned}$$

Choice (3) is the answer.

8.13. Based on the information given in the problem, we have:

$$f(x) = \begin{cases} x^2 + 1 & x \geq 1 \\ 2x + 3 & x < 1 \end{cases}$$

The problem can be solved as follows:

$$f(0) = 2 \times 0 + 3 = 3 \Rightarrow f(f(0)) = f(3) = 3^2 + 1 = 10$$

Choice (3) is the answer.

8.14. Based on the information given in the problem, we have:

$$f(x) = \sqrt{\frac{1 - |x|}{1 + |x|}}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Therefore:

$$\frac{1 - |x|}{1 + |x|} \geq 0 \xrightarrow{1 + |x| > 0} 1 - |x| \geq 0 \Rightarrow |x| \leq 1 \Rightarrow -1 \leq x \leq 1$$

Choice (4) is the answer.

8.15. Based on the information given in the problem, we have:

$$f(x) = \sqrt{\frac{x-1}{x-3}} + \sqrt{\frac{2-x}{x}}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Therefore:

$$\begin{cases} \frac{x-1}{x-3} \geq 0 \\ \frac{2-x}{x} \geq 0 \end{cases} \Rightarrow \begin{cases} \frac{x-1}{x-3} \geq 0 \\ \frac{x-2}{x} \leq 0 \end{cases} \Rightarrow \begin{cases} x \leq 1, x > 3 \\ 0 < x \leq 2 \end{cases} \Rightarrow 0 < x \leq 1$$

Note, that $x = 0$ must be excluded from the domain, since it makes the denominator zero. **Choice (1) is the answer.**

8.16. Based on the information given in the problem, we have:

$$f(x) = \frac{\sqrt{x}}{|x| - 1}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Moreover, the value of those x that make the denominator zero must be excluded from the domain. Thus:

$$\begin{cases} x \geq 0 \\ |x| - 1 \neq 0 \end{cases} \Rightarrow \begin{cases} x \geq 0 \\ |x| \neq 1 \end{cases} \Rightarrow \begin{cases} x \geq 0 \\ x \neq \pm 1 \end{cases} \Rightarrow D_f = [0, \infty) - \{1\}$$

Choice (2) is the answer.

8.17. Based on the information given in the problem, we have:

$$f(x) = \frac{\sqrt{x} + 1}{x\sqrt{x} + 1}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Moreover, the value of those x that make the denominator zero must be excluded from the domain. Thus:

$$\begin{cases} x \geq 0 \\ x\sqrt{x} + 1 \neq 0 \end{cases}$$

Note that $x\sqrt{x} + 1 \neq 0$ is true for any x . Hence:

$$\Rightarrow x \geq 0$$

Choice (3) is the answer.

8.18. Based on the information given in the problem, we have:

$$f(x) = \frac{1 - x}{4x + x^3}$$

The value of those x that make the denominator zero are not in the domain. Thus:

$$4x + x^3 \neq 0 \Rightarrow x(4 + x^2) \neq 0$$

Note that $4 + x^2 \neq 0$ for any x . Hence:

$$\Rightarrow x \neq 0$$

Choice (4) is the answer.

8.19. Based on the information given in the problem, we have:

$$f(x) = 2x \tag{1}$$

$$f(g(x)) = 2x + 2 \tag{2}$$

Therefore:

$$f(g(x)) = 2g(x) \quad (3)$$

Solving (2) and (3):

$$2g(x) = 2x + 2 \Rightarrow g(x) = x + 1$$

Choice (3) is the answer.

8.20. Based on the information given in the problem, we have:

$$f(x) = x - 1 \quad (1)$$

$$f(g(x)) = x \quad (2)$$

Thus:

$$f(g(x)) = g(x) - 1 \quad (3)$$

Solving (2) and (3):

$$g(x) - 1 = x \Rightarrow g(x) = x + 1$$

Choice (1) is the answer.

8.21. Based on the information given in the problem, we have:

$$f(x) = \sqrt{1-x}$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$y = \sqrt{1-x} \Rightarrow y^2 = 1-x \Rightarrow x = 1-y^2 \Rightarrow y = 1-x^2 \stackrel{\text{or}}{\Rightarrow} f^{-1}(x) = 1-x^2$$

Since the domain of $f^{-1}(x)$ is the same as the range of $f(x)$, which is $[0, \infty)$, we need to add $x \geq 0$ to $f^{-1}(x)$ as its domain. Thus:

$$f^{-1}(x) = 1-x^2, x \geq 0$$

Choice (1) is the answer.

8.22. Based on the information given in the problem, we have:

$$f(x) = \sin(x) - 2$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$y = \sin(x) - 2 \Rightarrow y + 2 = \sin(x) \Rightarrow x = \arcsin(y + 2) \Rightarrow y = \arcsin(\sin(x + 2)) \stackrel{\text{or}}{\Rightarrow} f^{-1}(x) = \arcsin(\sin(x + 2))$$

Choice (4) is the answer.

8.23. Based on the information given in the problem, we have:

$$f(x) = 3x - 2$$

$$g(x) = 2 + x$$

First, we need to determine $fog(x)$ as follows:

$$fog(x) = f(g(x)) = f(2 + x) = 3(2 + x) - 2 = 3x + 4$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$y = 3x + 4 \Rightarrow 3x = y - 4 \Rightarrow x = \frac{1}{3}y - \frac{4}{3} \Rightarrow y = \frac{1}{3}x - \frac{4}{3}$$

Choice (1) is the answer.

8.24. Based on the information given in the problem, we have:

$$f(x) = x^3 + 3x^2 + 3x + 2$$

First, we need to define the function in a cube form as follows:

$$\Rightarrow f(x) = (x^3 + 3x^2 + 3x + 1) + 1 = (x + 1)^3 + 1$$

To determine the inverse function of a function, it is convenient to use y instead of $f(x)$. Then, we need to determine x based on y . After that, we must replace x with y and vice versa. Note that the domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

Therefore:

$$\Rightarrow y = (x + 1)^3 + 1 \Rightarrow y - 1 = (x + 1)^3 \Rightarrow (y - 1)^{\frac{1}{3}} = x + 1 \Rightarrow x = (y - 1)^{\frac{1}{3}} - 1 \Rightarrow y = (x - 1)^{\frac{1}{3}} - 1$$

$$\stackrel{\text{or}}{\Rightarrow} f^{-1}(x) = -1 + \sqrt[3]{x - 1}$$

Choice (3) is the answer.

8.25. Based on the information given in the problem, we have:

$$f(x) + xf(-x) = x^2 + 1$$

The problem can be solved as follows:

$$\begin{aligned}
 \begin{matrix} \underline{\underline{x=2}} \\ \underline{\underline{x=-2}} \end{matrix} & \Rightarrow \begin{cases} f(2) + 2f(-2) = 2^2 + 1 \\ f(-2) + (-2)f(2) = (-2)^2 + 1 \end{cases} \Rightarrow \begin{cases} f(2) + 2f(-2) = 5 \\ f(-2) - 2f(2) = 5 \end{cases} \begin{matrix} \Rightarrow \\ \times(-2) \end{matrix} \Rightarrow \begin{cases} f(2) + 2f(-2) = 5 \\ -2f(-2) + 4f(2) = -10 \end{cases} \\
 & \Rightarrow 5f(2) = -5 \Rightarrow f(2) = -1
 \end{aligned}$$

Choice (1) is the answer.

8.26. From trigonometry, we know that:

$$1 + \cos(2x) = 2 \cos^2(x)$$

Based on the information given in the problem, we have:

$$f(x) = x^2 - 2$$

The problem can be solved as follows:

$$\begin{aligned}
 \Rightarrow f(2 \cos(x)) &= (2 \cos(x))^2 - 2 = 4 \cos^2(x) - 2 = 2(1 + \cos(2x)) - 2 = 2 \cos(2x) \\
 \Rightarrow f(f(2 \cos(x))) &= (2 \cos(2x))^2 - 2 = 4 \cos^2(2x) - 2 = 2(1 + \cos(4x)) - 2 = 2 \cos(4x) \\
 \Rightarrow f(f(f(2 \cos(x)))) &= (2 \cos(4x))^2 - 2 = 4 \cos^2(4x) - 2 = 2(1 + \cos(8x)) - 2 = 2 \cos(8x)
 \end{aligned}$$

Choice (4) is the answer.

8.27. Based on the information given in the problem, we have:

$$f\left(\frac{x-1}{x}\right) = \sqrt{2x-1}$$

First, we need to determine the $f(x)$ as follows:

$$\frac{x-1}{x} \equiv t \Rightarrow x = \frac{1}{1-t} \Rightarrow f(t) = \sqrt{2 \times \frac{1}{1-t} - 1} = \sqrt{\frac{1+t}{1-t}} \xrightarrow{t \rightarrow x} f(x) = \sqrt{\frac{1+x}{1-x}}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Therefore:

$$\frac{1+x}{1-x} \geq 0 \Rightarrow \frac{x+1}{x-1} \leq 0 \Rightarrow -1 \leq x < 1$$

Note, that $x = 1$ must be excluded from the domain, since it makes the denominator zero. Choice (3) is the answer.

8.28. Based on the information given in the problem, we have:

$$f(x) = \sqrt{1 - \sqrt{x-1}}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Therefore:

$$\begin{cases} 1 - \sqrt{x-1} \geq 0 \\ x-1 \geq 0 \end{cases} \Rightarrow \begin{cases} \sqrt{x-1} \leq 1 \\ x \geq 1 \end{cases} \Rightarrow \begin{cases} x-1 \leq 1 \\ x \geq 1 \end{cases} \Rightarrow \begin{cases} x \leq 2 \\ x \geq 1 \end{cases} \Rightarrow 1 \leq x \leq 2$$

Choice (2) is the answer.

8.29. Based on the information given in the problem, we have:

$$f(x) = \sqrt{|x| - 1} + \sqrt{|x| + 1}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero:

$$\begin{cases} |x| - 1 \geq 0 \\ |x| + 1 \geq 0 \end{cases} \Rightarrow \begin{cases} |x| \geq 1 \\ x \in \mathbb{R} \end{cases} \Rightarrow \begin{cases} x \leq -1, x \geq 1 \\ x \in \mathbb{R} \end{cases} \Rightarrow x \leq -1, x \geq 1 \Rightarrow D_f = \mathbb{R} - (-1, 1)$$

Note that $|x| + 1 \geq 0$ is true for any x . Choice (4) is the answer.

8.30. Based on the information given in the problem, we have:

$$f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$$

The problem can be solved as follows:

$$\begin{aligned} \Rightarrow f\left(x + \frac{1}{x}\right) &= x^2 + \frac{1}{x^2} + 2 - 2 = \left(x + \frac{1}{x}\right)^2 - 2 \\ x + \frac{1}{x} &\rightarrow x \\ \Rightarrow f(x) &= x^2 - 2 \Rightarrow f(3) = 3^2 - 2 = 7 \end{aligned}$$

Choice (3) is the answer.

8.31. Based on the information given in the problem, we have:

$$f(x) = |x + 2| + a|x - 2| \quad (1)$$

Based on the definition, the function of $f(x)$ is an even function if its domain is symmetric and:

$$f(-x) = f(x) \quad (2)$$

Therefore:

$$\Rightarrow f(-x) = |-x + 2| + a|-x - 2| = |-(x - 2)| + a|-(x + 2)| = |x - 2| + a|x + 2| \quad (3)$$

Solving (1), (2), and (3):

$$|x - 2| + a|x + 2| = |x + 2| + a|x - 2| \Rightarrow a = 1$$

Choice (3) is the answer.

8.32. Based on the information given in the problem, we have:

$$f(x) = x^2 + (A - 1)x \text{ is an even function} \quad (1)$$

$$g(x) = (B + 2)x^2 + \sin(x) \text{ is an odd function} \quad (2)$$

Based on the definition, the function of $f(x)$ is an even function if its domain is symmetric and:

$$f(-x) = f(x) \quad (3)$$

Additionally, the function of $f(x)$ is an odd function if its domain is symmetric and:

$$f(-x) = -f(x) \quad (4)$$

Solving (1) and (3):

$$(-x)^2 + (A - 1)(-x) = x^2 + (A - 1)x \Rightarrow 2(A - 1)x = 0 \Rightarrow A = 1$$

Solving (2) and (4):

$$\begin{aligned} (B + 2)(-x)^2 + \sin(-x) &= -((B + 2)x^2 + \sin(x)) \\ \Rightarrow (B + 2)x^2 - \sin(x) &= -(B + 2)x^2 - \sin(x) \Rightarrow 2(B + 2)x^2 = 0 \Rightarrow B = -2 \end{aligned}$$

Therefore:

$$A + B = 1 + (-2) = -1$$

Choice (2) is the answer.

8.33. Based on the definition, the function of $f(x)$ is an odd function if its domain is symmetric and:

$$f(-x) = -f(x)$$

Choice (1):

$$f(x) = \arccos(\cos(x)) \Rightarrow f(-x) = \arccos(\cos(-x)) = \pi - \arccos(\cos(x)) \neq \{-f(x), f(x)\} \Rightarrow \text{Not even nor odd}$$

Choice (2):

$$\begin{aligned} f(x) &= \sqrt{1-x} - \sqrt{1+x} \Rightarrow f(-x) = \sqrt{1-(-x)} - \sqrt{1+(-x)} = \sqrt{1+x} - \sqrt{1-x} \\ &= -(\sqrt{1-x} - \sqrt{1+x}) = -f(x) \Rightarrow \text{Odd} \end{aligned}$$

Choice (3):

$$f(x) = x^4 + x \Rightarrow f(-x) = (-x)^4 + (-x) = x^4 - x \neq \{-f(x), f(x)\} \Rightarrow \text{Not even nor odd}$$

Choice (4):

$$f(x) = x \sin(x) \Rightarrow f(-x) = -x \sin(-x) = x \sin(x) = f(x) \Rightarrow \text{Even}$$

Choice (2) is the answer.

8.34. Based on the definition, the function of $f(x)$ is an odd function if its domain is symmetric and:

$$f(-x) = -f(x)$$

Choice (1):

$$\begin{aligned} f(x) &= |x-1| + |x+1| \Rightarrow f(-x) = |-x-1| + |-x+1| = |-(x+1)| + |-(x-1)| \\ &= |x+1| + |x-1| = f(x) \Rightarrow \text{Even} \end{aligned}$$

Choice (2):

$$f(x) = \sin(|x|) \Rightarrow f(-x) = \sin(|-x|) = \sin(|x|) = f(x) \Rightarrow \text{Even}$$

Choice (3):

$$f(x) = x^3 + x^2 \Rightarrow f(-x) = (-x)^3 + (-x)^2 = -x^3 + x^2 \neq \{-f(x), f(x)\} \Rightarrow \text{Not even nor odd}$$

Choice (4):

$$\begin{aligned} f(x) &= |x-1| - |x+1| \Rightarrow f(-x) = |-x-1| - |-x+1| = |-(x+1)| - |-(x-1)| \\ &= |x+1| - |x-1| = -f(x) \Rightarrow \text{Odd} \end{aligned}$$

Choice (4) is the answer.

8.35. Based on the definition, the function of $f(x)$ is an even function if its domain is symmetric and:

$$f(-x) = f(x)$$

Choice (1):

$$\begin{aligned} f(x) &= |x-1| + |x+1| + |x| \Rightarrow f(-x) = |-x-1| + |-x+1| + |-x| \\ &= |-(x+1)| + |-(x-1)| + |-x| = |x+1| + |x-1| + |x| = f(x) \Rightarrow \text{Even} \end{aligned}$$

Choice (2):

$$f(x) = (x+1)^4 \Rightarrow f(-x) = (-x+1)^4 = (x-1)^4 \neq \{-f(x), f(x)\} \Rightarrow \text{Not even nor odd}$$

Choice (3):

$$f^2(x) + \sqrt[3]{x-1} = 0 \Rightarrow \text{Not a function}$$

Choice (4):

$$f(x) = [x] + 1 \Rightarrow f(-x) = [-x] + 1 \neq \{-f(x), f(x)\} \Rightarrow \text{Not even nor odd}$$

Choice (1) is the answer.

8.36. Based on the information given in the problem, we have:

$$g(x) = x - \frac{1}{x} \quad (1)$$

$$f(g(x)) = x^2 + \frac{1}{x^2} - 4 \quad (2)$$

Therefore:

$$\Rightarrow f(g(x)) = x^2 + \frac{1}{x^2} - 2 - 2 = \left(x - \frac{1}{x}\right)^2 - 2 \quad (3)$$

Solving (1) and (3):

$$f(g(x)) = (g(x))^2 - 2 \Rightarrow f(x) = x^2 - 2$$

Choice (2) is the answer.

8.37. Based on the information given in the problem, we have:

$$f(x) = \begin{cases} x^2 + 1 & x > 0 \\ 1 & x \leq 0 \end{cases}$$

As can be noticed from $f(x)$, the value of function is always positive. Therefore, the value of $-f(x)$ is always negative. Hence:

$$f(-f(x)) = 1$$

Choice (1) is the answer.

8.38. Based on the information given in the problem, we have:

$$f(x) = \sqrt{\log\left(\frac{5x - x^2}{4}\right)}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. Moreover, the domain of a logarithmic function can be determined by considering its argument greater than zero. Therefore:

$$\begin{aligned} \begin{cases} \log\left(\frac{5x - x^2}{4}\right) \geq 0 \\ \frac{5x - x^2}{4} > 0 \end{cases} &\Rightarrow \begin{cases} \log\left(\frac{5x - x^2}{4}\right) \geq \log(1) \\ x^2 - 5x < 0 \end{cases} \Rightarrow \begin{cases} \frac{5x - x^2}{4} \geq 1 \\ x(x - 5) < 0 \end{cases} \Rightarrow \begin{cases} x^2 - 5x + 4 \leq 0 \\ x(x - 5) < 0 \end{cases} \\ &\Rightarrow \begin{cases} (x - 4)(x - 1) \leq 0 \\ x(x - 5) < 0 \end{cases} \Rightarrow \begin{cases} 1 \leq x \leq 4 \\ 0 < x < 5 \end{cases} \Rightarrow 1 \leq x \leq 4 \end{aligned}$$

Choice (3) is the answer.

8.39. Based on the information given in the problem, we have:

$$f(x) = \sqrt{\log_x(x^2 + 9)}$$

The domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero. In addition, the domain of a logarithmic function can be determined by considering its argument greater than zero. In addition, the base of the logarithm must be greater than zero but not equal to one. Therefore:

$$\begin{cases} \log_x(x^2 + 9) \geq 0 \\ x^2 + 9 > 0 \\ x > 0, x \neq 1 \end{cases} \Rightarrow \begin{cases} \log_x(x^2 + 9) \geq \log_x(1) \\ x^2 + 9 > 0 \\ x > 0, x \neq 1 \end{cases} \Rightarrow \begin{cases} x^2 + 9 \geq 1 \\ x^2 + 9 > 0 \\ x > 0, x \neq 1 \end{cases}$$

Note that $x^2 + 9 > 0$ and $x^2 + 8 \geq 0$ are true for any x . Hence:

$$\Rightarrow x > 0, x \neq 1$$

Choice (4) is the answer.

8.40. Based on the information given in the problem, we have:

$$f(x) = 2x - 2[x] + 1$$

Based on the definition, we know that:

$$\begin{aligned} [x] \leq x < [x] + 1 &\xRightarrow{-[x]} 0 \leq x - [x] < 1 \xRightarrow{\times 2} 0 \leq 2x - 2[x] < 2 \xRightarrow{+1} 1 \leq 2x - 2[x] + 1 < 3 \\ &\Rightarrow 1 \leq f(x) < 3 \Rightarrow R_f = [1, 3) \end{aligned}$$

Choice (2) is the answer.

8.41. Based on the information given in the problem, we have:

$$f(x) = x^2 + 1$$

$$g(x) = \sqrt{x-1}$$

$$\Rightarrow fog(x) = f(g(x)) = f(\sqrt{x-1}) = (\sqrt{x-1})^2 + 1 = x - 1 + 1 = x$$

Next, we need to determine the domain of the function. As we know, the domain of a function in radical form, including even root, is determined by considering the radicand equal and greater than zero:

$$x - 1 \geq 0 \Rightarrow x \geq 1 \Rightarrow D_{fog} = [1, \infty)$$

Now, we can determine the range of the function based its domain as follows:

$$fog(x) = x \xRightarrow{D_{fog} = [1, \infty)} R_{fog} = [1, \infty)$$

Choice (2) is the answer.

8.42. Based on the information given in the problem, we have:

$$f(x) = \sqrt{x^2 - 2x + 3}$$

The problem can be solved as follows:

$$\Rightarrow f(x) = \sqrt{(x-1)^2 + 2}$$

As we know:

$$(x-1)^2 \geq 0 \xrightarrow{+2} (x-1)^2 + 2 \geq 2 \xrightarrow{\sqrt{}} \sqrt{(x-1)^2 + 2} \geq \sqrt{2} \Rightarrow f(x) \geq \sqrt{2} \Rightarrow R_{f(x)} = [\sqrt{2}, \infty)$$

Choice (1) is the answer.

8.43. Based on the information given in the problem, we have:

$$f(x) = |x-2| \Rightarrow D_f = \mathbb{R}$$

Based on the definition, two functions are equivalent if they are equal and their domains are the same.

Choice (1):

$$g_1(x) = \left| \frac{x^2 - 3x + 2}{x-1} \right| = \left| \frac{(x-1)(x-2)}{x-1} \right| = |x-2|$$

$$\Rightarrow D_{g_1} = \mathbb{R} - \{1\}$$

Therefore, the functions are not equivalent, since their domains are different. Note that $x = 1$ makes the denominator zero; thus, it is not in the domain.

Choice (2):

$$g_2(x) = \left| \frac{x^2 - 4}{x+2} \right| = \left| \frac{(x-2)(x+2)}{x+2} \right| = |x-2|$$

$$\Rightarrow D_{g_2} = \mathbb{R} - \{-2\}$$

Therefore, the functions are not equivalent because their domains are not the same. Note that $x = -2$ makes the denominator zero; thus, it must be excluded from the domain.

Choice (3):

$$g_3(x) = \frac{(x-2)^2}{|x-2|} = \frac{|x-2|^2}{|x-2|} = |x-2|$$

$$\Rightarrow D_{g_3} = \mathbb{R} - \{-2\}$$

Therefore, the functions are not equivalent because their domains are not the same. Note that $x = -2$ makes the denominator zero; thus, it must be excluded from the domain.

Choice (4):

$$g_4(x) = \frac{|6x-12|}{6} = \frac{6|x-2|}{6} = |x-2|$$

Problems: Factorization of Polynomials

9

Abstract

In this chapter, the basic and advanced problems of factorization of polynomials are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

9.1. Which one of the following terms is not a factor of $x^3 - 7x^2 + 6x$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) x
- 2) $x - 1$
- 3) $x - 6$
- 4) $x + 6$

9.2. Calculate the value of the following term:

$$\frac{x^3 + 8}{x^2 - 2x + 4}$$

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x + 2$
- 2) $x - 2$
- 3) $x + 1$
- 4) $x - 1$

9.3. Which one of the choices is a factor of $x^3 + y^3$?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x - y$
- 2) $x^2 + xy + y^2$
- 3) $x^2 - xy + y^2$
- 4) $x + y + xy$

9.4. Which one of the choices is a factor of $x^3 - y^3$?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x^2 + xy + y^2$
- 2) $x^2 - xy + y^2$
- 3) $x + y$
- 4) $x + y + xy$

9.5. Which one of the choices is not a factor of $x^3 + 2x^2 - 3x$?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x + 1$
- 2) x
- 3) $x - 1$
- 4) $x + 3$

9.6. Which one of the choices is a factor of $6x^3 + 7x^2 + 2x$?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $3x + 2$
- 2) $2x - 1$
- 3) $3x^2 - 2x$
- 4) $2x^2 - x$

9.7. Which one of the choices is a factor of $x^3 + x - 10$?

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x^2 - 2x + 5$
- 2) $x^2 + 2x + 5$
- 3) $x + 2$
- 4) $x^2 - 1$

9.8. Which one of the choices is a factor of $3a^2 + 2ab - b^2$?

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $a - 2b$
- 2) $3a - b$
- 3) $a - b$
- 4) $2a - b$

9.9. Which one of the choices is not a factor of $x^5 - x^4 - 4x + 4$?

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $x^2 - 2$
- 2) $x + 1$
- 3) $x^2 + 2$
- 4) $x - 1$

9.10. Calculate the value of $x^2 + \frac{1}{x^2}$ if $x + \frac{1}{x} = 2$.

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 8
- 2) 6
- 3) 4
- 4) 2

9.11. Calculate the value of the following term:

$$\frac{(x+1)^3 - 3x(x+1)}{x^3 + 1}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $\frac{x^3+1-3x^2-3x}{x^3+1}$

2) 1

3) $\frac{x^3-3x}{x^3+1}$

4) $\frac{x^3-3x^2}{x^3+1}$

9.12. Calculate the value of $(50.01)^2 - (49.99)^2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) 0.2

2) 0.02

3) 2

4) 0

9.13. Which one of the following terms is not a factor of $x^4 + 2x^3 - x - 2$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $x^2 - x + 1$

2) $x^2 + x + 1$

3) $x + 2$

4) $x - 1$

9.14. Which one of the following terms shows the correct factorization of $x^4 + x^2 - 2$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $(x^2 - 2)(x^2 + 1)$

2) $(x^2 + 2)(x^2 + 1)$

3) $(x^2 + 2)(x + 1)(x - 1)$

4) $(x^2 - 2)(x + 1)(x - 1)$

9.15. Calculate the value of the term below:

$$\frac{x^3 - 2x^{\frac{3}{2}} + 1}{(x^{\frac{3}{2}} - 1)^2}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) 2

2) $x - 2$

3) 1

4) x

9.16. Calculate the value of $a^3 - b^3$ if $a - b = 1$ and $a^2 + b^2 = 5$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) 10

2) 9

3) 7

4) 6

9.17. Calculate the value of the following term:

$$\frac{(a+b)^2 + (b+c)^2 + (a+c)^2 - (a+b+c)^2}{a^2 + b^2 + c^2}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) 1

2) $\frac{ab+bc+ac}{a^2+b^2+c^2}$

3) $\frac{(a+b+c)^2}{a^2+b^2+c^2}$

4) $\frac{2(ab+bc+ac)}{a^2+b^2+c^2}$

9.18. Calculate the value of $x^3 + y^3$ if $xy = 5$ and $x + y = 7$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) 216

2) 238

3) 244

4) 264

9.19. Calculate the value of the term below:

$$\frac{2x^4 + x^3 - 4x - 2}{x^3 - 2}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) $2x$ 2) $2x - 1$ 3) $2x + 1$ 4) $x + 1$

9.20. Calculate the value of the term below:

$$\frac{x^4 + x^2 + 1}{x^2 + x + 1}$$

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) $x^2 + x + 1$ 2) $x^2 + x$ 3) $x^2 - x + 1$ 4) $x^2 + x - 1$

9.21. Calculate the value of the following term:

$$\frac{x^6 + 4x^2 + 5}{x^2 + 1}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

1) $x^4 + x^2 + 5$ 2) $x^4 - x^2 + 5$

- 3) $x^4 - 2x^2 + 5$
 4) $x^4 + 2x^2 + 5$

9.22. Calculate the value of the following term:

$$\frac{x^9 - 1}{x^3 + x^3 + 1}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $x^3 + 1$
 2) $x^3 - 1$
 3) $x^3 + 2$
 4) $x^3 - 2$

9.23. Calculate the value of $x^2 + y^2$ if $x^4 + y^4 + 4 = 2(2x^2 + 2y^2 - x^2y^2)$.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 2
 2) 1
 3) 0
 4) 8

9.24. Calculate the value of the following term:

$$(2 + 1)(2^2 + 1)(2^4 + 1) \dots (2^{64} + 1)$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $2^{256} + 1$
 2) $2^{128} + 1$
 3) $2^{128} - 1$
 4) $2^{256} - 1$

Solutions of Problems: Factorization of Polynomials

10

Abstract

In this chapter, the problems of the ninth chapter are fully solved, in detail, step-by-step, and with different methods.

10.1. The problem can be solved as follows:

$$x^3 - 7x^2 + 6x = x(x^2 - 7x + 6) = x(x - 1)(x - 6)$$

As can be seen, $x + 6$ is not a factor of the expression. **Choice (4) is the answer.**

10.2. Based on the rule of sum of two cubes, we know that:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Therefore:

$$\frac{x^3 + 8}{x^2 - 2x + 4} = \frac{(x + 2)(x^2 - 2x + 4)}{x^2 - 2x + 4} = x + 2$$

Choice (1) is the answer.

10.3. Based on the rule of sum of two cubes, we know that:

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

Choice (3) is the answer.

10.4. Based on the rule of difference of two cubes, we know that:

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

Choice (1) is the answer.

10.5. The problem can be solved as follows:

$$x^3 + 2x^2 - 3x = x(x^2 + 2x - 3) = x(x + 3)(x - 1)$$

Choice (1) is the answer.

10.6. The problem can be solved as follows:

$$6x^3 + 7x^2 + 2x = x(6x^2 + 7x + 2) = x(3x + 2)(2x + 1)$$

Choice (1) is the answer.

10.7. Based on the rule of difference of two cubes, we know that:

$$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$$

Therefore:

$$x^3 + x - 10 = (x^3 - 8) + (x - 2) = (x - 2)(x^2 + 2x + 4) + (x - 2) = (x - 2)(x^2 + 2x + 5)$$

Choice (2) is the answer.

10.8. Based on the rule of difference of two squares, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

The problem can be solved as follows:

$$3a^2 + 2ab - b^2 = 2a^2 + 2ab + a^2 - b^2 = 2a(a + b) + (a + b)(a - b) = (a + b)(2a + a - b) = (a + b)(3a - b)$$

Choice (2) is the answer.

10.9. From the rule of difference of two squares, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

The problem can be solved as follows:

$$x^5 - x^4 - 4x + 4 = x^5 - 4x - (x^4 - 4) = x(x^4 - 4) - (x^4 - 4) = (x^4 - 4)(x - 1) = (x^2 + 2)(x^2 - 2)(x - 1)$$

Choice (2) is the answer.

10.10. Based on the information given in the problem, we have:

$$x + \frac{1}{x} = 2 \tag{1}$$

The problem can be solved as follows:

$$x^2 + \frac{1}{x^2} = x^2 + \frac{1}{x^2} + 2 - 2 = \left(x + \frac{1}{x}\right)^2 - 2 \tag{2}$$

Solving (1) and (2):

$$x^2 + \frac{1}{x^2} = (2)^2 - 2 = 2$$

Choice (4) is the answer.

10.11. As we know:

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Therefore:

$$\frac{(x+1)^3 - 3x(x+1)}{x^3 + 1} = \frac{x^3 + 3x^2 + 3x + 1 - 3x^2 - 3x}{x^3 + 1} = \frac{x^3 + 1}{x^3 + 1} = 1$$

Choice (2) is the answer.

10.12. Based on the rule of difference of two squares, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

The problem can be solved as follows:

$$(50.01)^2 - (49.99)^2 = (50.01 + 49.99)(50.01 - 49.99) = (100)(0.02) = 2$$

Choice (3) is the answer.

10.13. Based on the rule of difference of two cubes, we know that:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

The problem can be solved as follows:

$$x^4 + 2x^3 - x - 2 = x^3(x + 2) - (x + 2) = (x + 2)(x^3 - 1) = (x + 2)(x - 1)(x^2 + x + 1)$$

Choice (1) is the answer.

10.14. Based on the rule of difference of two squares, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

The problem can be solved as follows:

$$x^4 + x^2 - 2 = x^4 - 1 + x^2 - 1 = (x^2 - 1)(x^2 + 1) + x^2 - 1 = (x^2 - 1)(x^2 + 2) = (x - 1)(x + 1)(x^2 + 2)$$

Choice (3) is the answer.

10.15. Based on perfect square binomial formula, we have:

$$a^2 + b^2 - 2ab = (a - b)^2$$

The problem can be solved as follows:

$$\frac{x^3 - 2x^{\frac{3}{2}} + 1}{(x^{\frac{3}{2}} - 1)^2} = \frac{\left(x^{\frac{3}{2}}\right)^2 - 2x^{\frac{3}{2}} + 1}{(x^{\frac{3}{2}} - 1)^2} = \frac{\left(x^{\frac{3}{2}} - 1\right)^2}{(x^{\frac{3}{2}} - 1)^2} = 1$$

Choice (3) is the answer.

10.16. From the rule of difference of two cubes, we know that:

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab) \quad (1)$$

Based on the information given in the problem, we have:

$$a - b = 1 \quad (2)$$

$$a^2 + b^2 = 5 \quad (3)$$

From (2), we have:

$$a - b = 1 \Rightarrow (a - b)^2 = 1 \Rightarrow a^2 + b^2 - 2ab = 1 \quad (4)$$

Solving (3) and (4):

$$5 - 2ab = 1 \Rightarrow ab = 2 \quad (5)$$

Solving (1), (2), (3), and (5):

$$a^3 - b^3 = 1 \times (5 + 2) = 7$$

Choice (3) is the answer.

10.17. As we know:

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$$

Therefore:

$$\begin{aligned} & \frac{(a + b)^2 + (b + c)^2 + (a + c)^2 - (a + b + c)^2}{a^2 + b^2 + c^2} \\ &= \frac{a^2 + b^2 + 2ab + b^2 + c^2 + 2bc + a^2 + c^2 + 2ac - (a^2 + b^2 + c^2 + 2ab + 2bc + 2ac)}{a^2 + b^2 + c^2} \\ &= \frac{a^2 + b^2 + c^2}{a^2 + b^2 + c^2} = 1 \end{aligned}$$

Choice (1) is the answer.

10.18. Based on the information given in the problem, we have:

$$xy = 5$$

$$x + y = 7$$

Based on the rule of sum of two cubes, we know that:

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$$

Therefore:

$$x^3 + y^3 = (x + y)(x^2 + 2xy + y^2 - 3xy) = (x + y)((x + y)^2 - 3xy) = 7 \times (7^2 - 3 \times 5) = 7 \times 34 = 238$$

Choice (2) is the answer.

10.19. The problem can be solved as follows:

$$\frac{2x^4 + x^3 - 4x - 2}{x^3 - 2} = \frac{(2x^4 - 4x) + (x^3 - 2)}{x^3 - 2} = \frac{2x(x^3 - 2) + (x^3 - 2)}{x^3 - 2} = \frac{(x^3 - 2)(2x + 1)}{x^3 - 2} = 2x + 1$$

Choice (3) is the answer.

10.20. Based on the rule of difference of two squares and perfect square binomial formula, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

$$a^2 + b^2 + 2ab = (a + b)^2$$

The problem can be solved as follows:

$$\begin{aligned} \frac{x^4 + x^2 + 1}{x^2 + x + 1} &= \frac{x^4 + x^2 + 1 + x^2 - x^2}{x^2 + x + 1} = \frac{x^4 + 2x^2 + 1 - x^2}{x^2 + x + 1} = \frac{(x^2 + 1)^2 - x^2}{x^2 + x + 1} \\ &= \frac{(x^2 + 1 + x)(x^2 + 1 - x)}{x^2 + x + 1} = x^2 - x + 1 \end{aligned}$$

Choice (3) is the answer.

10.21. Based on the information given in the problem, the answer of the following term is requested:

$$\frac{x^6 + 4x^2 + 5}{x^2 + 1}$$

The problem can be solved as follows:

$$\begin{array}{r|l} x^6 + 4x^2 + 5 & x^2 + 1 \\ \underline{x^6 + x^4} & x^4 - x^2 + 5 \\ -x^4 + 4x^2 & \\ \underline{-x^4 - x^2} & \\ 5x^2 + 5 & \\ \underline{5x^2 + 5} & \\ 0 & \end{array}$$

Choice (2) is the answer.

10.22. Based on the rule of difference of two cubes, we know that:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Therefore:

$$\frac{x^9 - 1}{x^3 + x^{\frac{3}{2}} + 1} = \frac{(x^{\frac{3}{2}} - 1)\left((x^{\frac{3}{2}})^2 + x^{\frac{3}{2}} + 1\right)}{(x^{\frac{3}{2}})^2 + x^{\frac{3}{2}} + 1} = x^{\frac{3}{2}} - 1$$

Choice (2) is the answer.

10.23. As we know:

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ac) \quad (1)$$

Therefore:

$$\begin{aligned} x^4 + y^4 + 4 &= 2(2x^2 + 2y^2 - x^2y^2) \Rightarrow x^4 + y^4 + 4 - 2(2x^2 + 2y^2 - x^2y^2) = 0 \\ \Rightarrow (-x^2)^2 + (-y^2)^2 + 2^2 + 2(2(-x^2) + 2(-y^2) + (-x^2)(-y^2)) &= 0 \end{aligned} \quad (2)$$

Solving (1) and (2):

$$((-x^2) + (-y^2) + 2)^2 = 0 \Rightarrow -x^2 - y^2 + 2 = 0 \Rightarrow x^2 + y^2 = 2$$

Choice (1) is the answer.

10.24. Based on the rule of difference of two squares, we know that:

$$(a + b)(a - b) = a^2 - b^2$$

Now, the problem can be solved as follows:

$$\begin{aligned} &(2 + 1)(2^2 + 1)(2^4 + 1) \dots (2^{64} + 1) \\ \xrightarrow{\times(2-1)} &(2 - 1)(2 + 1)(2^2 + 1)(2^4 + 1) \dots (2^{64} + 1) = (2^2 - 1)(2^2 + 1)(2^4 + 1) \dots (2^{64} + 1) \\ &= (2^4 - 1)(2^4 + 1) \dots (2^{64} + 1) = (2^8 - 1) \dots (2^{64} + 1) = \dots = (2^{64} - 1)(2^{64} + 1) = 2^{128} - 1 \end{aligned}$$

Choice (3) is the answer.

Problems: Trigonometric and Inverse Trigonometric Functions

11

Abstract

In this chapter, the basic and advanced problems of trigonometric and inverse trigonometric functions are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

- 11.1. Calculate the value of $2 \cos(x)$ if $\sin(x) = \frac{1}{2}$ and the end of arc x is in the second quadrant.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $-\sqrt{3}$
- 2) $-\sqrt{2}$
- 3) $\sqrt{2}$
- 4) $\sqrt{3}$

- 11.2. If $\cos(\alpha)(1 + \tan^2(\alpha)) > 0$, then determine the quadrant that the end of arc α is located in that.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) First
- 2) Second
- 3) First or third
- 4) First or fourth

- 11.3. Calculate the value of $\cos(10^\circ) + \cos(190^\circ)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 0
- 2) $2 \cos(10^\circ)$
- 3) $2 \sin(10^\circ)$
- 4) $\cos(200^\circ)$

- 11.4. Calculate the value of $\sin(200^\circ) + \sin(10^\circ) + \sin(20^\circ) + \sin(190^\circ)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $2 \sin(10^\circ)$
- 2) $2 \cos(10^\circ)$
- 3) 0
- 4) $-\sin(10^\circ)$

11.5. Calculate the value of $\sin(1860^\circ) + \cos(1860^\circ)$

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{1+\sqrt{3}}{2}$
- 2) $\frac{\sqrt{3}-1}{2}$
- 3) $\frac{1-\sqrt{3}}{2}$
- 4) 0

11.6. Calculate the value of $\tan(135^\circ)\sin^2(60^\circ)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $-\frac{3}{4}$
- 2) $-\frac{1}{2}$
- 3) $\frac{1}{4}$
- 4) $\frac{3}{4}$

11.7. Calculate the value of $\cos^2\left(\frac{5\pi}{4}\right) + \sin\left(\frac{7\pi}{6}\right)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 1
- 2) $\frac{1}{2}$
- 3) $\frac{1}{4}$
- 4) 0

11.8. Calculate the value of $(\sin(60^\circ) - \sin(45^\circ))(\cos(30^\circ) + \cos(45^\circ))$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{3}{4}$
- 2) $\frac{1}{4}$
- 3) $\frac{1}{2}$
- 4) 1

11.9. If $\cos(\theta) = -\frac{2}{3}$ and $\tan(\theta)\cos(\theta) > 0$, which quadrant is the location of the end of arc θ ?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) First
- 2) Second
- 3) Third
- 4) Fourth

11.10. In Figure 11.1, in triangle ABC, determine the value of $\sin(\theta)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{3}{5}$
- 2) $\frac{3}{4}$
- 3) $\frac{4}{5}$
- 4) $\frac{5}{4}$

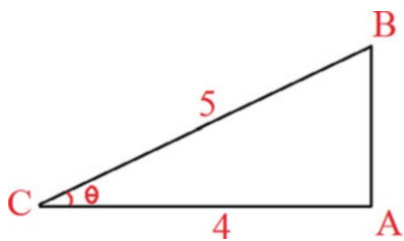


Figure 11.1 The graph of problem 11.10

11.11. In Figure 11.2, determine the value of $\tan(\alpha) + \tan(\beta)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{5}{6}$
- 2) $\frac{4}{5}$

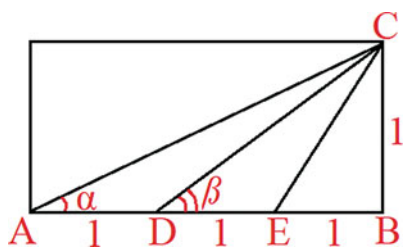


Figure 11.2 The graph of problem 11.11

- 3) $\frac{6}{5}$
- 4) $\frac{5}{4}$

11.12. Calculate the value of $\cos(60^\circ) \cos(30^\circ) + \sin(60^\circ) \sin(30^\circ)$.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\tan(30^\circ)$
- 2) $\cot(45^\circ)$
- 3) $\sin(60^\circ)$
- 4) $\cos(60^\circ)$

11.13. Which one of the statements below is correct?

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) $\sin(50^\circ) < \sin(40^\circ)$
- 2) $\cos(50^\circ) < \cos(40^\circ)$
- 3) $\tan(50^\circ) < \tan(40^\circ)$
- 4) $\cot(40^\circ) < \cot(50^\circ)$

11.14. Simplify and calculate the final answer of the term below:

$$\frac{\sin(60^\circ) \cos(60^\circ) \tan(60^\circ) \csc(60^\circ) \sec(60^\circ) \cot(60^\circ)}{\sin(40^\circ) \cos(40^\circ) \tan(40^\circ) \csc(40^\circ) \sec(40^\circ) \cot(40^\circ)}$$

Difficulty level ○ Easy ● Normal ○ Hard
Calculation amount ○ Small ● Normal ○ Large

- 1) 0
- 2) $\frac{1}{2}$
- 3) 1

4) 2

11.15. Calculate the value of $\sin(135^\circ) + \cos(45^\circ) + \tan(225^\circ) + \cot(315^\circ)$.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $2 + \sqrt{2}$
 2) $\sqrt{2}$
 3) $\sqrt{2} - 2$
 4) 2

11.16. Which one of the choices is equal to $\tan(10^\circ)$?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\tan(-10^\circ)$
 2) $\cot(100^\circ)$
 3) $\tan(170^\circ)$
 4) $\tan(190^\circ)$

11.17. Calculate the value of $\tan(\theta)$ if $\sin(\theta) = -\frac{\sqrt{5}}{5}$ and the end of arc θ is in the third quadrant.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) -2
 2) $-\frac{1}{2}$
 3) $\frac{1}{2}$
 4) 2

11.18. Which one of the choices is equivalent to the angle of -27° ?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 127°
 2) -127°
 3) 333°
 4) 53°

11.19. Calculate the range of m if:

$$\cos(3x) = \frac{m-1}{2}, \quad -\frac{\pi}{9} < x < \frac{\pi}{9}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) (1, 2]
 2) (0, 2)
 3) (2, 3]
 4) (2, 3)

11.20. Calculate the range of m if:

$$\sin(x) = \frac{m}{2}, \quad \frac{\pi}{4} < x < \frac{3\pi}{5}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $m \leq \frac{\sqrt{2}}{2}$
 2) $m > \frac{\sqrt{2}}{2}$

3) $\sqrt{2} < m \leq 2$

4) $1 \leq m < \sqrt{2}$

11.21. Calculate the range of m if:

$$\cos(2\alpha) = \frac{1}{1-m}, \quad \frac{\pi}{4} < \alpha < \frac{3\pi}{4}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) $(-\infty, 2)$

2) $(1, \infty)$

3) $[2, \infty)$

4) $(-\infty, 1)$

11.22. Calculate the range of m if:

$$\sin(x) = \frac{3-m^2}{3+m^2}, \quad \frac{\pi}{3} < x < \frac{5\pi}{6}$$

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) $|m| < \sqrt{3}$

2) $|m| < \sqrt{2}$

3) $|m| < 1$

4) $|m| < \frac{1}{2}$

11.23. If $\sin(\alpha) \cos(\alpha) > 0$ and $\cos(\alpha) \cot(\alpha) < 0$, then determine the quadrant that the end of arc α is located.

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) First

2) Second

3) Third

4) Fourth

11.24. Which one of the inequalities is correct?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

1) $\sin(20^\circ) > \sin(170^\circ)$

2) $\cos(20^\circ) < \cos(160^\circ)$

3) $\sin(20^\circ) < \sin(10^\circ)$

4) $\cos(10^\circ) < \cos(20^\circ)$

Solutions of Problems: Trigonometric and Inverse Trigonometric Functions 12

Abstract

In this chapter, the problems of the 11th chapter are fully solved, in detail, step-by-step, and with different methods.

12.1. Based on the information given in the problem, we know that:

$$\frac{\pi}{2} \leq x \leq \pi$$

$$\sin(x) = \frac{1}{2}$$

Moreover, from trigonometry, we know that:

$$\sin^2(x) + \cos^2(x) = 1$$

The problem can be solved as follows:

$$2 \cos(x) = 2\sqrt{1 - \sin^2(x)} = 2\sqrt{1 - \left(\frac{1}{2}\right)^2} = 2\sqrt{\frac{3}{4}} = \pm\sqrt{3}$$

As can be noticed from Figure 12.1, $\sqrt{3}$ is not acceptable for $\cos(x)$ because the end of arc x is in the second quadrant ($\cos(x) < 0$). Therefore:

$$2 \cos(x) = -\sqrt{3}$$

Choice (1) is the answer.

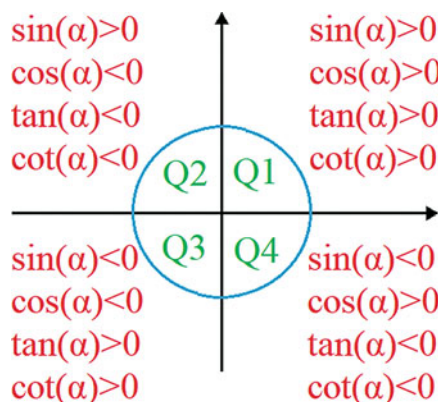


Figure 12.1 The graph of the solution of problem 12.1

12.2. Based on the information given in the problem, we know that:

$$\cos(\alpha)(1 + \tan^2(\alpha)) > 0 \xrightarrow{1 + \tan^2(\alpha) > 0} \cos(\alpha) > 0$$

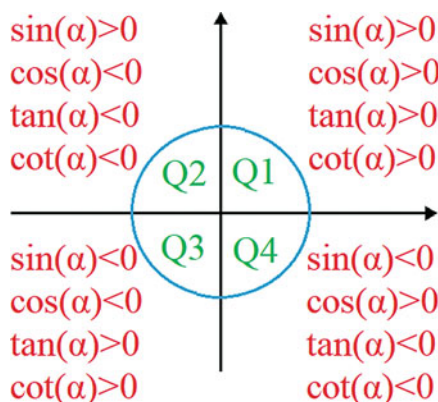


Figure 12.2 The graph of the solution of problem 12.2

Therefore, the end of arc α is in the first or fourth quadrant, as can be noticed from Figure 12.2. **Choice (4) is the answer.**

12.3. From trigonometry, we know that for an acute angle of θ :

$$\cos(180^\circ + \theta) = -\cos(\theta)$$

Therefore:

$$\cos(10^\circ) + \cos(190^\circ) = \cos(10^\circ) + \cos(180^\circ + 10^\circ) = \cos(10^\circ) - \cos(10^\circ) = 0$$

Choice (1) is the answer.

12.4. From trigonometry, we know that for an acute angle of θ :

$$\sin(180^\circ + \theta) = -\sin(\theta)$$

Therefore:

$$\sin(200^\circ) + \sin(10^\circ) + \sin(20^\circ) + \sin(190^\circ)$$

$$= \sin(180^\circ + 20^\circ) + \sin(10^\circ) + \sin(20^\circ) + \sin(180^\circ + 10^\circ) = -\sin(20^\circ) + \sin(10^\circ) + \sin(20^\circ) - \sin(10^\circ) = 0$$

Choice (3) is the answer.

12.5. From trigonometry, we know that for an acute angle of θ :

$$\sin(n \times 360^\circ + \theta) = \sin(\theta), n \in \mathbb{Z}$$

$$\cos(n \times 360^\circ + \theta) = \cos(\theta), n \in \mathbb{Z}$$

Moreover, we know that:

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}$$

$$\cos(60^\circ) = \frac{1}{2}$$

Therefore:

$$\begin{aligned}\sin(1860^\circ) + \cos(1860^\circ) &= \sin(5 \times 360^\circ + 60^\circ) + \cos(5 \times 360^\circ + 60^\circ) \\ &= \sin(60^\circ) + \cos(60^\circ) = \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3} + 1}{2}\end{aligned}$$

Choice (1) is the answer.

12.6. From trigonometry, we know that for an acute angle of θ :

$$\tan(180^\circ - \theta) = -\tan(\theta)$$

In addition, we know that:

$$\tan(45^\circ) = 1$$

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}$$

Therefore:

$$\begin{aligned}\tan(135^\circ) \sin^2(60^\circ) &= \tan(135^\circ) \sin^2(60^\circ) = \tan(180^\circ - 45^\circ) \sin^2(60^\circ) = -\tan(45^\circ) \sin^2(60^\circ) = -1 \times \left(\frac{\sqrt{3}}{2}\right)^2 \\ &= -\frac{3}{4}\end{aligned}$$

Choice (1) is the answer.

12.7. From trigonometry, we know that for an acute angle of θ :

$$\cos(\pi + \theta) = -\cos(\theta)$$

$$\sin(\pi + \theta) = -\sin(\theta)$$

Moreover, we know that:

$$\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

Therefore:

$$\cos^2\left(\frac{5\pi}{4}\right) + \sin\left(\frac{7\pi}{6}\right) = \cos^2\left(\pi + \frac{\pi}{4}\right) + \sin\left(\pi + \frac{\pi}{6}\right) = \left(-\cos\left(\frac{\pi}{4}\right)\right)^2 - \sin\left(\frac{\pi}{6}\right) = \left(-\frac{\sqrt{2}}{2}\right)^2 - \frac{1}{2} = \frac{1}{2} - \frac{1}{2} = 0$$

Choice (4) is the answer.

12.8. From trigonometry, we know that:

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}$$

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}$$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}$$

$$\cos(45^\circ) = \frac{\sqrt{2}}{2}$$

In addition, based on the rule of difference of two squares, we know that:

$$(a+b)(a-b) = a^2 - b^2$$

The problem can be solved as follows:

$$(\sin(60^\circ) - \sin(45^\circ))(\cos(30^\circ) + \cos(45^\circ)) = \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\right) = \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{3}{4} - \frac{2}{4} = \frac{1}{4}$$

Choice (2) is the answer.

12.9. Based on the information given in the problem, we know that:

$$\cos(\theta) = -\frac{2}{3} \Rightarrow \cos(\theta) < 0 \quad (1)$$

$$\tan(\theta) \cos(\theta) > 0 \quad (2)$$

Solving (1) and (2):

$$\tan(\theta) < 0 \quad (3)$$

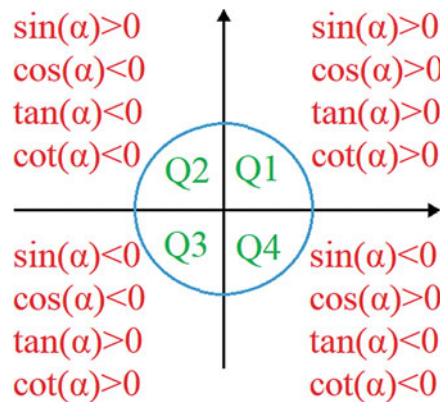


Figure 12.3 The graph of the solution of problem 12.9

By considering Figure 12.3 as well as (1) and (3), we can conclude that the location of the end of arc θ is in the second quadrant. Choice (2) is the answer.

12.10. From trigonometry, we know that:

$$\sin(\text{angle}) = \frac{\text{Side opposite to the angle}}{\text{Hypotenuse}}$$

Moreover, by using Pythagorean formula for the triangle shown in Figure 12.4, we can write:

$$BC^2 = AB^2 + AC^2 \Rightarrow AB = \sqrt{BC^2 - AC^2} = \sqrt{5^2 - 4^2} = 3$$

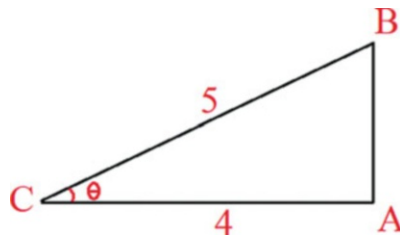


Figure 12.4 The graph of the solution of problem 12.10

Therefore:

$$\sin(\theta) = \frac{AB}{BC} = \frac{3}{5}$$

Choice (1) is the answer.

12.11. From trigonometry, we know that:

$$\tan(\text{angle}) = \frac{\text{Side opposite to the angle}}{\text{Side adjacent to the angle}}$$

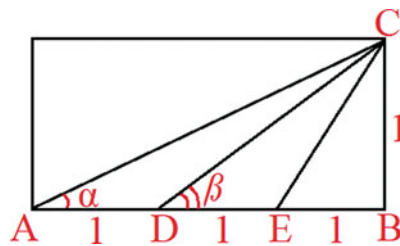


Figure 12.5 The graph of the solution of problem 12.11

Therefore:

$$\tan(\alpha) + \tan(\beta) = \frac{CB}{AB} + \frac{CB}{DB} = \frac{1}{1+1+1} + \frac{1}{1+1} = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

Choice (1) is the answer.

12.12. From trigonometry, we know that:

$$\cos(60^\circ) = \frac{1}{2}$$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2}$$

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}$$

$$\sin(30^\circ) = \frac{1}{2}$$

Therefore:

$$\cos(60^\circ) \cos(30^\circ) + \sin(60^\circ) \sin(30^\circ) = \frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2} = \sin(60^\circ)$$

Choice (3) is the answer.

12.13. We know that for an acute angle, $\sin(\alpha)$ and $\tan(\alpha)$ have ascending trends. Therefore:

$$\sin(40^\circ) < \sin(50^\circ)$$

$$\tan(40^\circ) < \tan(50^\circ)$$

Moreover, for an acute angle, $\cos(\alpha)$ and $\cot(\alpha)$ are descending. Therefore:

$$\cos(50^\circ) < \cos(40^\circ)$$

$$\cot(50^\circ) < \cot(40^\circ)$$

Choice (2) is the answer.

12.14. From trigonometry, we know that:

$$\cot(\alpha) = \frac{1}{\tan(\alpha)}$$

$$\sec(\alpha) = \frac{1}{\cos(\alpha)}$$

$$\csc(\alpha) = \frac{1}{\sin(\alpha)}$$

The problem can be solved as follows:

$$\begin{aligned} & \frac{\sin(60^\circ) \cos(60^\circ) \tan(60^\circ) \csc(60^\circ) \sec(60^\circ) \cot(60^\circ)}{\sin(40^\circ) \cos(40^\circ) \tan(40^\circ) \csc(40^\circ) \sec(40^\circ) \cot(40^\circ)} \\ &= \frac{\sin(60^\circ) \csc(60^\circ) \cos(60^\circ) \sec(60^\circ) \tan(60^\circ) \cot(60^\circ)}{\sin(40^\circ) \csc(40^\circ) \cos(40^\circ) \sec(40^\circ) \tan(40^\circ) \cot(40^\circ)} = \\ &= \frac{\sin(60^\circ) \frac{1}{\sin(60^\circ)} \cos(60^\circ) \frac{1}{\cos(60^\circ)} \tan(60^\circ) \frac{1}{\tan(60^\circ)}}{\sin(40^\circ) \frac{1}{\sin(40^\circ)} \cos(40^\circ) \frac{1}{\cos(40^\circ)} \tan(40^\circ) \frac{1}{\tan(40^\circ)}} = \frac{1 \times 1 \times 1}{1 \times 1 \times 1} = 1 \end{aligned}$$

Choice (3) is the answer.

12.15. From trigonometry, we know that for an acute angle of θ :

$$\sin(180^\circ - \theta) = \sin(\theta)$$

$$\tan(180^\circ + \theta) = \tan(\theta)$$

$$\cot(360^\circ - \theta) = -\cot(\theta)$$

In addition, we know that:

$$\sin(45^\circ) = \frac{\sqrt{2}}{2}$$

$$\cos(45^\circ) = \frac{\sqrt{2}}{2}$$

$$\tan(45^\circ) = 1$$

$$\cot(45^\circ) = 1$$

The problem can be solved as follows:

$$\begin{aligned} \sin(135^\circ) + \cos(45^\circ) + \tan(225^\circ) + \cot(315^\circ) &= \sin(180^\circ - 45^\circ) + \cos(45^\circ) + \tan(180^\circ + 45^\circ) \\ &\quad + \cot(360^\circ - 45^\circ) \\ &= \sin(45^\circ) + \cos(45^\circ) + \tan(45^\circ) - \cot(45^\circ) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} + 1 - 1 = \sqrt{2} \end{aligned}$$

Choice (2) is the answer.

12.16. From trigonometry, we know that for an acute angle of θ :

$$\tan(-\theta) = -\tan(\theta)$$

$$\cot(90^\circ + \theta) = -\tan(\theta)$$

$$\tan(180^\circ - \theta) = -\tan(\theta)$$

$$\tan(180^\circ + \theta) = \tan(\theta)$$

Choice (1):

$$\tan(-10^\circ) = -\tan(10^\circ)$$

Choice (2):

$$\cot(100^\circ) = \cot(90^\circ + 10^\circ) = -\tan(10^\circ)$$

Choice (3):

$$\tan(170^\circ) = \tan(180^\circ - 10^\circ) = -\tan(10^\circ)$$

Choice (4):

$$\tan (190^{\circ}) = \tan (180^{\circ} + 10^{\circ}) = \tan (10^{\circ})$$

Choice (4) is the answer.

12.17. Based on the information given in the problem, we know that the end of arc θ is in the third quadrant and:

$$\sin (\theta) = -\frac{\sqrt{5}}{5} \quad (1)$$

$$\sin^2(\theta) + \cos^2(\theta) = 1 \quad (2)$$

$$\tan (\theta) = \frac{\sin (\theta)}{\cos (\theta)} \quad (3)$$

The problem can be solved as follows:

Solving (1) and (2):

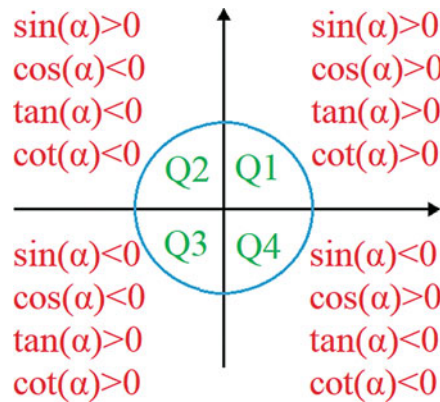


Figure 12.6 The graph of the solution of problem 12.17

$$\cos (\theta) = \sqrt{1 - \sin^2(\theta)} = \sqrt{1 - \left(\frac{\sqrt{5}}{5}\right)^2} = \sqrt{1 - \frac{1}{5}} = \pm \frac{2}{\sqrt{5}} \quad (4)$$

In (4), the negative value is acceptable because the end of arc θ is in the third quadrant (see Figure 12.6).

Solving (1), (3), and (4):

$$\tan (\theta) = \frac{\sin (\theta)}{\cos (\theta)} = \frac{-\frac{\sqrt{5}}{5}}{-\frac{2}{\sqrt{5}}} = \frac{1}{2}$$

Choice (3) is the answer.

12.18. As we know from trigonometry, in a unit circle, the relation below holds for any pair of angles:

$$\theta_2 - \theta_1 = n \times 360^{\circ}, n \in \mathbb{Z} \Rightarrow \theta_2 \equiv \theta_1$$

The problem can be solved as follows:

Choice (1):

$$127^\circ - (-27^\circ) = 154^\circ \Rightarrow 127^\circ \neq -27^\circ$$

Choice (2):

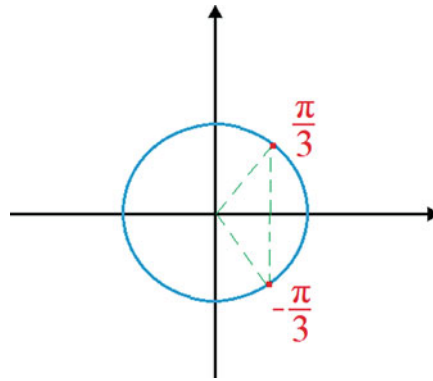


Figure 12.7 The graph of the solution of problem 12.19

$$-127^\circ - (-27^\circ) = 100^\circ \Rightarrow -127^\circ \neq -27^\circ$$

Choice (3):

$$333^\circ - (-27^\circ) = 360^\circ \Rightarrow 333^\circ \equiv -27^\circ$$

Choice (4):

$$53^\circ - (-27^\circ) = 80^\circ \Rightarrow 53^\circ \neq -27^\circ$$

Choice (3) is the answer.

12.19. Calculate the range of m if:

$$\cos(3x) = \frac{m-1}{2}, \quad -\frac{\pi}{9} < x < \frac{\pi}{9}$$

The problem can be solved as follows:

$$-\frac{\pi}{9} < x < \frac{\pi}{9} \xrightarrow{\times 3} -\frac{\pi}{3} < 3x < \frac{\pi}{3} \quad (1)$$

Equation (1) and Figure 12.7 show the range of $3x$. Therefore:

$$\Rightarrow \frac{1}{2} < \cos(3x) \leq 1 \Rightarrow \frac{1}{2} < \frac{m-1}{2} \leq 1 \Rightarrow 1 < m-1 \leq 2 \Rightarrow 2 < m \leq 3$$

Choice (3) is the answer.

12.20. Calculate the range of m if:

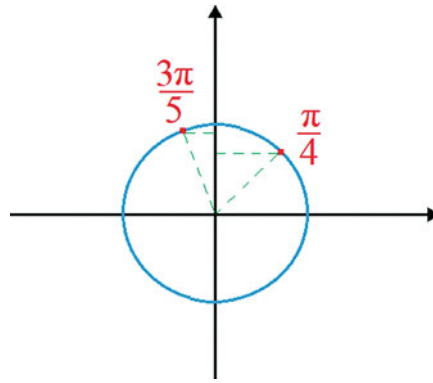


Figure 12.8 The graph of the solution of problem 12.20

$$\sin(x) = \frac{m}{2} \quad (1)$$

$$\frac{\pi}{4} < x < \frac{3\pi}{5} \quad (2)$$

Equation (2) and Figure 12.8 show the range of x . Therefore:

$$\Rightarrow \frac{\sqrt{2}}{2} < \sin(x) \leq 1 \Rightarrow \frac{\sqrt{2}}{2} < \frac{m}{2} \leq 1 \Rightarrow \sqrt{2} < m \leq 2$$

Choice (3) is the answer.

12.21. Based on the information given in the problem, we have:

$$\cos(2\alpha) = \frac{1}{1-m}, \quad \frac{\pi}{4} < \alpha < \frac{3\pi}{4}$$

The problem can be solved as follows:

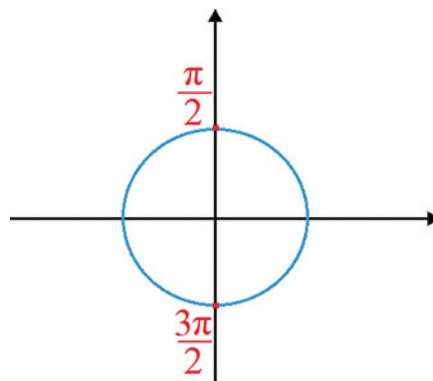


Figure 12.9 The graph of the solution of problem 12.21

$$\frac{\pi}{4} < \alpha < \frac{3\pi}{4} \xrightarrow{\times 2} \frac{\pi}{2} < 2\alpha < \frac{3\pi}{2} \quad (1)$$

As can be seen in (1) and Figure 12.9, 2α is in the second and third quadrants. Therefore:

$$\Rightarrow -1 \leq \cos(2\alpha) < 0 \Rightarrow -1 \leq \frac{1}{1-m} < 0$$

$$\Rightarrow \begin{cases} \frac{1}{1-m} < 0 \Rightarrow 1-m < 0 \Rightarrow m > 1 \\ \frac{1}{1-m} \geq -1 \Rightarrow \frac{1}{1-m} + 1 \geq 0 \Rightarrow \frac{1+1-m}{1-m} \geq 0 \Rightarrow \frac{2-m}{1-m} \geq 0 \Rightarrow \frac{m-2}{m-1} \geq 0 \Rightarrow m < 1, m \geq 2 \end{cases} \Rightarrow m \geq 2$$

Choice (3) is the answer.

12.22. Calculate the range of m if:

$$\sin(x) = \frac{3-m^2}{3+m^2}, \quad \frac{\pi}{3} < x < \frac{5\pi}{6}$$

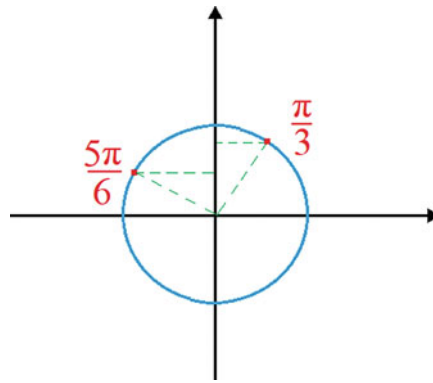


Figure 12.10 The graph of the solution of problem 12.22

The problem can be solved as follows:

$$\frac{\pi}{3} < x < \frac{5\pi}{6} \quad (1)$$

Equation (1) and Figure 12.10 show the range of x . Therefore:

$$\Rightarrow \frac{1}{2} \leq \sin(x) < 1 \Rightarrow \frac{1}{2} < \frac{3-m^2}{3+m^2} \leq 1$$

$$\Rightarrow \begin{cases} \frac{3-m^2}{3+m^2} \leq 1 \Rightarrow 3-m^2 \leq 3+m^2 \Rightarrow m^2 \geq 0 \Rightarrow m \in \mathbb{R} \\ \frac{1}{2} < \frac{3-m^2}{3+m^2} \Rightarrow 3+m^2 < 6-2m^2 \Rightarrow 3m^2 < 3 \Rightarrow m^2 < 1 \Rightarrow -1 < m < 1 \Rightarrow |m| < 1 \end{cases} \Rightarrow |m| < 1$$

Choice (3) is the answer.

12.23. From trigonometry, we know that:

$$\cot(\alpha) = \frac{\cos(\alpha)}{\sin(\alpha)} \quad (1)$$

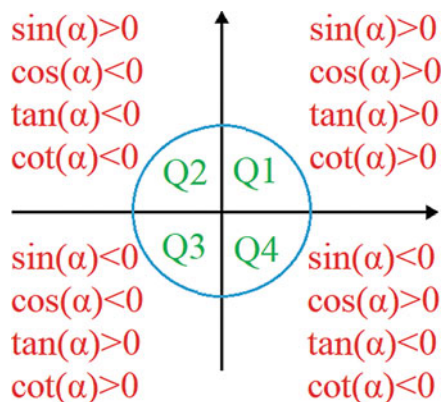


Figure 12.11 The graph of the solution of problem 12.23

Based on the information given in the problem, we know that:

$$\sin(\alpha) \cos(\alpha) > 0 \quad (2)$$

$$\cos(\alpha) \cot(\alpha) < 0 \quad (3)$$

Solving (1) and (3):

$$\cos(\alpha) \frac{\cos(\alpha)}{\sin(\alpha)} < 0 \Rightarrow \frac{\cos^2(\alpha)}{\sin(\alpha)} < 0 \xrightarrow{\cos^2(\alpha) > 0} \frac{1}{\sin(\alpha)} < 0 \Rightarrow \sin(\alpha) < 0 \quad (4)$$

Solving (2) and (4) and considering Figure 12.11:

$$\cos(\alpha) < 0 \quad (5)$$

Therefore, based on (4) and (5), the end of arc α is in the third quadrant. **Choice (3) is the answer.**

12.24. From trigonometry, we know that for an acute angle of θ :

$$\sin(180^\circ - \theta) = \sin(\theta)$$

$$\sin(180^\circ + \theta) = -\sin(\theta)$$

$$\cos(180^\circ - \theta) = -\cos(\theta)$$

$$\cos(180^\circ + \theta) = -\cos(\theta)$$

Moreover, we know that for an acute angle of θ , $\sin(\theta)$ and $\cos(\theta)$ have ascending and descending trends, respectively. Therefore:



Problems: Arithmetic and Geometric Sequences and Series

13

Abstract

In this chapter, the basic and advanced problems of arithmetic and geometric sequences and series are presented. To help students study the chapter in the most efficient way, the problems are categorized in different levels based on their difficulty levels (easy, normal, and hard) and calculation amounts (small, normal, and large). Moreover, the problems are ordered from the easiest problem with the smallest computations to the most difficult problems with the largest calculations.

- 13.1. If the 12th term and the common difference of successive terms of an arithmetic sequence are 34 and 2, respectively, determine its 18th term.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 46
- 2) 48
- 3) 50
- 4) 52

- 13.2. In an arithmetic sequence, the first term and the common difference of successive terms are 2 and 4, respectively. Determine the ratio of the 11th term to the first term in this arithmetic sequence.

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 19
- 2) 20
- 3) 21
- 4) 42

- 13.3. In the arithmetic sequence of $2, \frac{7}{3}, \dots$, which term is equal to 6?

Difficulty level ● Easy ○ Normal ○ Hard
Calculation amount ● Small ○ Normal ○ Large

- 1) 10
- 2) 11
- 3) 12
- 4) 13

- 13.4. The common term of a geometric sequence is presented in the following. Determine the common ratio of successive terms of this geometric sequence:

$$\frac{2}{3 \times 2^n}$$

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) $\frac{1}{6}$
- 2) $\frac{1}{3}$
- 3) $\frac{1}{2}$
- 4) $\frac{2}{3}$

- 13.5. In an arithmetic sequence, the first term and the common difference of successive terms are 0.5 and -1 , respectively. Calculate the first eight terms of this arithmetic sequence.

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) -16
- 2) -20
- 3) -24
- 4) -25

- 13.6. Which term of the arithmetic sequence of 2, 5, 8, ... is equal to 56?

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) 18
- 2) 19
- 3) 20
- 4) 21

- 13.7. In an arithmetic sequence, $a_1 = 4$ and $a_{n+1} = a_n + 3$. Determine the n -th term of the sequence.

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) $n + 5$
- 2) $3n + 1$
- 3) $2n + 3$
- 4) $4n - 1$

- 13.8. In an arithmetic sequence, the sum of the first and the third terms is equal to 1.5 times of the sum of the second and the fourth terms. What relation exists between the first term and the common difference of successive terms?

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) $a_1 = -4d$
- 2) $d = -4a_1$
- 3) $d = a_1$
- 4) $a_1 = -d$

- 13.9. Calculate the sum of the infinite number of terms of the geometric sequence of 12, 8, $\frac{16}{3}$, ...

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) 32
- 2) 36
- 3) 48
- 4) 54

- 13.10. Calculate the geometric mean of $\sqrt{15}$ and $\sqrt{240}$.

Difficulty level ● Easy ○ Normal ○ Hard
 Calculation amount ● Small ○ Normal ○ Large

- 1) $3\sqrt{13}$
- 2) $2\sqrt{13}$

3) $3\sqrt{15}$

4) $2\sqrt{15}$

- 13.11. Determine the fifth term of the geometric sequence of
- $4, -6, 9, \dots$
- .

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ● Small ○ Normal ○ Large

1) $-\frac{71}{4}$

2) $-\frac{51}{4}$

3) $\frac{81}{4}$

4) $\frac{61}{4}$

- 13.12. Calculate the sum of the first six terms of the geometric sequence of
- $32, 16, \dots$
- .

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ● Small ○ Normal ○ Large

1) 58

2) 62

3) 63

4) 66

- 13.13. Calculate the limiting sum of the infinite geometric sequence of
- $-12, 4, -\frac{4}{3}, \dots$
- .

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ● Small ○ Normal ○ Large

1) -3

2) -6

3) -9

4) -18

- 13.14. Calculate the geometric mean of
- $\sqrt{3}$
- and
- $\frac{\sqrt{3}}{4}$
- .

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ● Small ○ Normal ○ Large

1) $\frac{\sqrt{3}}{2}$

2) $\frac{3}{2}$

3) $\frac{\sqrt{3}}{4}$

4) $\frac{3}{4}$

- 13.15. Calculate the arithmetic mean of 3 and 5.

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ● Small ○ Normal ○ Large

1) 4

2) 3

3) 5

4) $\sqrt{15}$

- 13.16. If the second and the fifth terms of a geometric sequence are 6 and
- $\frac{16}{9}$
- , respectively, determine the common ratio of successive terms of the geometric sequence.

Difficulty level ● Easy ○ Normal ○ Hard

Calculation amount ○ Small ● Normal ○ Large

1) $\frac{2}{3}$

2) $\frac{3}{4}$

3) $\frac{4}{3}$

4) $\frac{3}{2}$

- 13.17. Determine the first term of an arithmetic sequence if the sum of its first 12 terms is 120 and the common difference of successive terms is 2.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) -2
 2) -1
 3) 1
 4) 2

- 13.18. In a geometric sequence, the sum of the first and the third terms is equal to 1.5 times of the sum of the second and the fourth terms. Determine the common ratio of the successive terms.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\frac{1}{3}$
 2) $\frac{1}{2}$
 3) $\frac{2}{3}$
 4) $\frac{3}{2}$

- 13.19. In a geometric sequence, the subtraction of the fifth and third terms is equal to $\frac{1}{32}$. If the common ratio of successive terms is $\frac{1}{2}$, determine the first term of the geometric sequence.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $-\frac{2}{3}$
 2) $-\frac{1}{6}$
 3) $\frac{1}{8}$
 4) $\frac{3}{2}$

- 13.20. Calculate the common ratio of successive terms of a descending geometric sequence that its limiting sum is equal to four times of the sum of the first two terms.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) $\frac{\sqrt{3}}{4}$
 2) $\frac{1}{4}$
 3) $\frac{\sqrt{3}}{2}$
 4) $\frac{3}{4}$

- 13.21. In a geometric sequence, the first and seventh terms are 10 and 640, respectively. Calculate the sum of the first six terms of the sequence.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 1260
 2) 640
 3) 730
 4) 630

- 13.22. Which one of the following successive terms belongs to an arithmetic sequence?

Difficulty level ☐ Easy ☒ Normal ☐ Hard
 Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{1}{\sqrt{5}}, 1, \sqrt{5}$
 2) $\frac{5}{3}, 1, \frac{2}{3}$
 3) $\frac{9}{5}, \frac{6}{5}, \frac{3}{5}$

4) $\frac{27}{5}, \frac{9}{5}, \frac{3}{5}$

- 13.23. Calculate the common difference of successive terms of an arithmetic sequence if its 7th and 15th terms are 20 and 30, respectively.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) $\frac{3}{5}$
2) $\frac{4}{5}$
3) $\frac{5}{4}$
4) $\frac{5}{3}$

- 13.24. The seventh and the ninth terms of an arithmetic sequence are 15 and 19, respectively. Calculate the 12th term of this arithmetic sequence.

Difficulty level ☐ Easy ☒ Normal ☐ Hard
Calculation amount ☐ Small ☒ Normal ☐ Large

- 1) 24
2) 25
3) 27
4) 29

- 13.25. The terms of $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are the successive terms of a geometric sequence. Which one of the following statements is true about the successive terms of $\log a, \log b, \log c$?

Difficulty level ☐ Easy ☐ Normal ☒ Hard
Calculation amount ☒ Small ☐ Normal ☐ Large

- 1) They are the successive terms of an arithmetic sequence.
2) They are the successive terms of a geometric sequence.
3) They can be the successive terms of an arithmetic or a geometric sequence.
4) They are not the successive terms of a sequence.

Solutions of Problems: Arithmetic and Geometric Sequences and Series 14

Abstract

In this chapter, the problems of the 13th chapter are fully solved, in detail, step-by-step, and with different methods.

14.1. As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (1)$$

Based on the information given in the problem, we have:

$$a_{12} = 34 \quad (2)$$

$$d = 2 \quad (3)$$

Solving (1), (2), and (3):

$$34 = a_1 + (12 - 1) \times 2 \Rightarrow a_1 = 12 \quad (4)$$

Solving (1), (3), and (4):

$$a_{18} = 12 + (18 - 1) \times 2 = 46$$

Choice (1) is the answer.

14.2. As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d$$

Based on the information given in the problem, we have:

$$a_1 = 2$$

$$d = 4$$

Therefore:

$$\frac{a_{11}}{a_1} = \frac{a_1 + (n - 1)d}{a_1} = \frac{2 + (11 - 1) \times 4}{2} = \frac{42}{2} = 21$$

Choice (3) is the answer.

14.3. From the arithmetic sequence of $2, \frac{7}{3}, \dots$, the following information can be extracted:

$$a_1 = 2 \quad (1)$$

$$d = \frac{7}{3} - 2 = \frac{1}{3} \quad (2)$$

Moreover, the information below is given in the problem:

$$a_n = 6 \quad (3)$$

As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (4)$$

Solving (1)–(4):

$$6 = 2 + (n - 1) \times \frac{1}{3} \Rightarrow (n - 1) \times \frac{1}{3} = 4 \Rightarrow n - 1 = 12 \Rightarrow n = 13$$

Choice (4) is the answer.

14.4. Based on the information given in the problem, we have:

$$a_n = \frac{2}{3 \times 2^n}$$

As we know, the common ratio of successive terms of a geometric sequence can be determined as follows:

$$q = \frac{a_{n+1}}{a_n}$$

Therefore:

$$q = \frac{a_2}{a_1} = \frac{\frac{2}{3 \times 2^2}}{\frac{2}{3 \times 2^1}} = \frac{\frac{1}{3 \times 2}}{\frac{1}{3}} = \frac{1}{2}$$

Choice (3) is the answer.

14.5. Based on the information given in the problem, we have:

$$a_1 = \frac{1}{2}$$

$$d = -1$$

As we know, the sum the first n terms of an arithmetic sequence can be calculated using the formula below:

$$S_n = \frac{n}{2}(2a_1 + (n - 1)d)$$

Therefore:

$$S_8 = \frac{8}{2} \left(2 \times \frac{1}{2} + (8 - 1) \times (-1) \right) = 4(1 - 7) = -24$$

Choice (3) is the answer.

14.6. As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (1)$$

By looking at the arithmetic sequence of 2, 5, 8, ..., the following information is achieved:

$$a_1 = 2 \quad (1)$$

$$d = 5 - 2 = 3 \quad (1)$$

Solving (1), (2), and (3) for $a_n = 56$:

$$56 = 2 + (n - 1) \times 3 \Rightarrow n - 1 = \frac{54}{3} = 18 \Rightarrow n = 19$$

Choice (2) is the answer.

14.7. As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (1)$$

Based on the information given in the problem, we have:

$$a_1 = 4 \quad (2)$$

$$a_{n+1} = a_n + 3 \quad (3)$$

The common difference of successive terms can be calculated as follows:

$$d = a_{n+1} - a_n = 3 \quad (4)$$

By solving (1), (2), and (4), the n -th term of the sequence can be determined:

$$a_n = 4 + (n - 1) \times 3 = 3n + 1$$

Choice (2) is the answer.

14.8. Based on the information given in the problem, we have:

$$a_1 + a_3 = 1.5(a_2 + a_4) \quad (1)$$

As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (2)$$

Solving (1) and (2):

$$a_1 + a_1 + (3 - 1)d = 1.5(a_1 + (2 - 1)d + a_1 + (4 - 1)d) \Rightarrow 2a_1 + 2d = 1.5(2a_1 + 4d)$$

$$\Rightarrow a_1 = -4d$$

Choice (1) is the answer.

14.9. By looking at the geometric sequence of $12, 8, \frac{16}{3}, \dots$, the following results can be achieved:

$$a_1 = 12$$

$$q = \frac{8}{12} = \frac{2}{3}$$

The sum of infinite number of terms of a geometric sequence, that the magnitude of its common ratio of successive terms is less than one, can be calculated as follows:

$$\lim_{n \rightarrow \infty} S_n = \frac{a_1}{1 - q}, \quad |q| < 1$$

Therefore:

$$\lim_{n \rightarrow \infty} S_n = \frac{a_1}{1 - q} = \frac{12}{1 - \frac{2}{3}} = 36$$

Choice (2) is the answer.

14.10. In a geometric sequence, the geometric mean ($G.M.$) of a and b can be calculated as follows:

$$G.M. = \sqrt{a \times b}$$

Therefore:

$$G.M. = \sqrt{\sqrt{15} \times \sqrt{240}} = \sqrt{\sqrt{15} \times \sqrt{16 \times 15}} = \sqrt{\sqrt{15} \times 4\sqrt{15}} = \sqrt{4 \times 15} = 2\sqrt{15}$$

Choice (4) is the answer.

14.11. As we know, the common term of a geometric sequence is as follows:

$$a_n = a_1 q^{n-1} \quad (1)$$

Moreover, the common ratio of successive terms of a geometric sequence can be determined as follows:

$$q = \frac{a_{n+1}}{a_n} \quad (2)$$

From the geometric sequence of $4, -6, 9, \dots$, the information below can be extracted:

$$a_1 = 4 \quad (3)$$

$$q = \frac{a_{n+1}}{a_n} = \frac{-6}{4} = -\frac{3}{2} \quad (4)$$

Solving (1), (3), and (4):

$$a_5 = 4\left(-\frac{3}{2}\right)^{5-1} = 4 \times \left(\frac{81}{16}\right) = \frac{81}{4}$$

Choice (3) is the answer.

14.12. From the geometric sequence of 32, 16, ..., the following results can be concluded:

$$a_1 = 32$$

$$q = \frac{16}{32} = \frac{1}{2}$$

The sum of finite number of terms of a geometric sequence can be calculated as follows:

$$S_n = \frac{a_1(1 - q^n)}{1 - q}$$

Therefore:

$$S_6 = \frac{32\left(1 - \left(\frac{1}{2}\right)^6\right)}{1 - \frac{1}{2}} = \frac{32\left(1 - \frac{1}{64}\right)}{\frac{1}{2}} = \frac{32 \times \frac{63}{64}}{\frac{1}{2}} = 63$$

Choice (3) is the answer.

14.13. By looking at the geometric sequence of $-12, 4, -\frac{4}{3}, \dots$, the following results are achieved:

$$a_1 = -12$$

$$q = \frac{4}{-12} = -\frac{1}{3}$$

The limiting sum of the infinite geometric sequence can be calculated as follows:

$$S = \frac{a_1}{1 - q}, \quad |q| < 1$$

Therefore:

$$S = \frac{a_1}{1 - q} = \frac{-12}{1 - \left(-\frac{1}{3}\right)} = \frac{-12}{\frac{4}{3}} = -9$$

Choice (3) is the answer.

14.14. In a geometric sequence, the geometric mean (*G. M.*) of a and b can be calculated as follows:

$$G. M. = \sqrt{a \times b}$$

Therefore:

$$G. M. = \sqrt{\sqrt{3} \times \frac{\sqrt{3}}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

Choice (1) is the answer.

14.15. In an arithmetic sequence, the arithmetic mean (A. M.) of a and b can be calculated as follows:

$$A. M. = \frac{a+b}{2}$$

Therefore:

$$A. M. = \frac{a+b}{2} = \frac{3+5}{2} = 4$$

Choice (1) is the answer.

14.16. Based on the information given in the problem, we have:

$$a_2 = 6 \tag{1}$$

$$a_5 = \frac{16}{9} \tag{2}$$

As we know, the common term of a geometric sequence is as follows:

$$a_n = a_1 q^{n-1} \tag{3}$$

Solving (1) and (3):

$$6 = a_1 q^{2-1} \Rightarrow 6 = a_1 q \tag{4}$$

Solving (2) and (3):

$$\frac{16}{9} = a_1 q^{5-1} \Rightarrow \frac{16}{9} = a_1 q^4 \tag{5}$$

Solving (4) and (5):

$$\frac{a_1 q^4}{a_1 q} = \frac{\frac{16}{9}}{6} \Rightarrow q^3 = \frac{8}{27} \Rightarrow q = \frac{2}{3}$$

Choice (1) is the answer.

14.17. As we know, the sum the first n terms of an arithmetic sequence can be calculated using the formula below:

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

Based on the information given in the problem, we know that:

$$n = 12$$

$$S_{12} = 120$$

$$d = 2$$

Therefore:

$$120 = \frac{12}{2}(2a_1 + (12 - 1) \times 2) \Rightarrow 20 = 2a_1 + 22 \Rightarrow -2 = 2a_1 \Rightarrow a_1 = -1$$

Choice (2) is the answer.

14.18. Based on the information given in the problem, we have:

$$a_1 + a_3 = 1.5(a_2 + a_4) \quad (1)$$

As we know, the common term of a geometric sequence is as follows:

$$a_n = a_1 q^{n-1} \quad (2)$$

Solving (1) and (2):

$$a_1 + a_1 q^{3-1} = 1.5(a_1 q^{2-1} + a_1 q^{4-1}) \Rightarrow 1 + q^2 = 1.5q(1 + q^2) \Rightarrow 1 = 1.5q \Rightarrow q = \frac{2}{3}$$

Choice (3) is the answer.

14.19. Based on the information given in the problem, we have:

$$a_5 - a_3 = \frac{1}{32} \quad (1)$$

$$q = \frac{1}{2} \quad (2)$$

As we know, the common term of a geometric sequence is as follows:

$$a_n = a_1 q^{n-1} \quad (3)$$

Solving (1)–(3):

$$a_1 \left(\frac{1}{2}\right)^{5-1} - a_1 \left(\frac{1}{2}\right)^{3-1} = \frac{1}{32} \Rightarrow a_1 \left(\left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^2\right) = \frac{1}{32} \Rightarrow a_1 = \frac{\frac{1}{32}}{\frac{1}{16} - \frac{1}{4}} = \frac{\frac{1}{32}}{-\frac{3}{16}} = -\frac{1}{6}$$

Choice (2) is the answer.

14.20. Based on the information given in the problem, we have:

$$S_\infty = 4(a_1 + a_2) \quad (1)$$

The common term of a geometric sequence is as follows:

$$a_n = a_1 q^{n-1} \quad (2)$$

In addition, the limiting sum of the infinite geometric sequence can be calculated as follows:

$$S = \frac{a_1}{1-q}, \quad |q| < 1 \quad (3)$$

Therefore:

$$\frac{a_1}{1-q} = 4(a_1 + a_1q) \Rightarrow \frac{1}{1-q} = 4(1+q) \Rightarrow 1-q^2 = \frac{1}{4} \Rightarrow q^2 = \frac{3}{4} \Rightarrow q = \frac{\sqrt{3}}{2}$$

Choice (3) is the answer.

14.21. Based on the information given in the problem, we have:

$$a_1 = 10 \quad (1)$$

$$a_7 = 640 \quad (2)$$

Moreover, the common term of a geometric sequence is as follows:

$$a_n = a_1q^{n-1} \quad (3)$$

The sum of finite number of terms of a geometric sequence can be calculated as follows:

$$S_n = \frac{a_1(1-q^n)}{1-q} \quad (4)$$

Solving (1), (2), and (3):

$$640 = 10q^{7-1} \Rightarrow q^6 = 64 \Rightarrow q = 2 \quad (5)$$

Solving (1), (4), and (5):

$$S_6 = \frac{10(1-2^6)}{1-2} = \frac{10(-63)}{-1} = 630$$

Choice (4) is the answer.

14.22. If three successive terms belong to an arithmetic sequence, the relation below, which is used to calculate the arithmetic mean of the successive terms of a_1 and a_3 , must be held:

$$a_2 = \frac{a_1 + a_3}{2}$$

Choice (1):

$$1 \neq \frac{\frac{1}{\sqrt{5}} + \sqrt{5}}{2} \Rightarrow 1 \neq \frac{3\sqrt{5}}{5}$$

Choice (2):

$$1 \neq \frac{\frac{5}{3} + \frac{2}{3}}{2} \Rightarrow 1 \neq \frac{7}{6}$$

Choice (3):

$$\frac{6}{5} = \frac{\frac{9}{5} + \frac{3}{5}}{2} \Rightarrow \frac{6}{5} = \frac{6}{5}$$

Choice (4):

$$\frac{9}{5} \neq \frac{\frac{27}{5} + \frac{3}{5}}{2} \Rightarrow \frac{9}{5} \neq 3$$

Choice (3) is the answer.

14.23. As we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (1)$$

Based on the information given in the problem, we have:

$$a_7 = 20 \quad (2)$$

$$a_{15} = 30 \quad (3)$$

Solving (1), (2), and (3):

$$\begin{cases} 20 = a_1 + (7 - 1)d \\ 30 = a_1 + (15 - 1)d \end{cases} \Rightarrow \begin{cases} 20 = a_1 + 6d \\ 30 = a_1 + 14d \end{cases} \xrightarrow{-} 10 = 8d \Rightarrow d = \frac{5}{4}$$

Choice (3) is the answer.

14.24. Based on the information given in the problem, we have:

$$a_7 = 15 \quad (1)$$

$$a_9 = 19 \quad (2)$$

Moreover, as we know, the common term of an arithmetic sequence is as follows:

$$a_n = a_1 + (n - 1)d \quad (3)$$

Therefore:

$$\begin{cases} 15 = a_1 + (7 - 1)d \\ 19 = a_1 + (9 - 1)d \end{cases} \Rightarrow \begin{cases} 15 = a_1 + 6d \\ 19 = a_1 + 8d \end{cases} \xrightarrow{-} 4 = 2d \Rightarrow d = 2 \quad (4)$$

$$15 = a_1 + 6d \xrightarrow{d=2} 15 = a_1 + 12 \Rightarrow a_1 = 3 \quad (5)$$

Solving (3), (4), and (5) for $n = 12$:

$$a_{12} = 3 + (12 - 1) \times 2 = 3 + 22 = 25$$

Choice (2) is the answer.

14.25. In a geometric sequence, the relation below is held, where a_2 is the geometric mean of the successive terms of a_1, a_2, a_3 if:

$$(a_2)^2 = a_1 a_3 \quad (1)$$

Moreover, in an arithmetic sequence, the relation below is held, where a_2 is the arithmetic mean of the successive terms of a_1, a_2, a_3 if:

$$a_2 = \frac{a_1 + a_3}{2} \quad (2)$$

In addition, we know that the relations below are held for logarithm:

$$\log(x^2) = 2 \log(x) \quad (3)$$

$$\log(xy) = \log(x) + \log(y) \quad (4)$$

Based on the information given in the problem, we know that the terms of $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are the successive terms of a geometric sequence. Therefore, by considering (1), we can write:

$$\left(\frac{1}{b}\right)^2 = \frac{1}{a} \times \frac{1}{c} \Rightarrow b^2 = ac \quad (5)$$

$$\stackrel{\text{Log}}{\Rightarrow} \log(b^2) = \log(ac) \Rightarrow 2 \log(b) = \log(a) + \log(c) \Rightarrow \log(b) = \frac{\log(a) + \log(c)}{2}$$

Therefore, by considering (2), we see that the successive terms of $\log a, \log b, \log c$ belong to an arithmetic sequence.

Choice (1) is the answer.

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